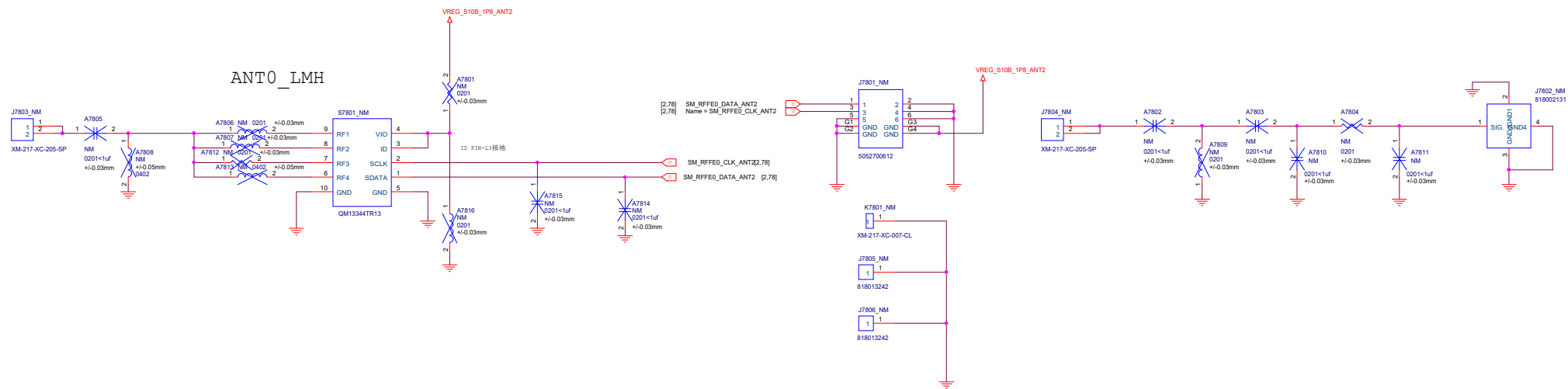
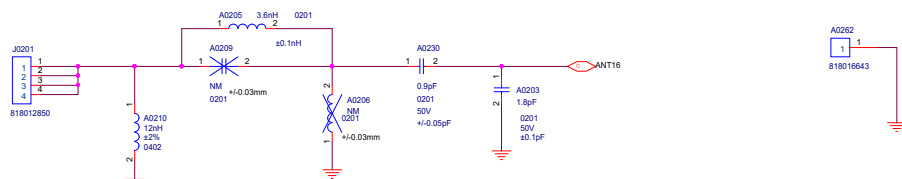


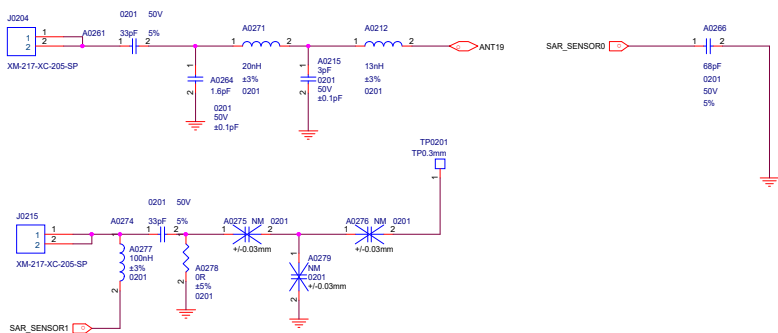
[illegible]



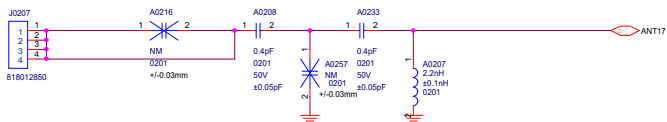
ANT16 GPS L1+WIFI2.4 CHAIN0



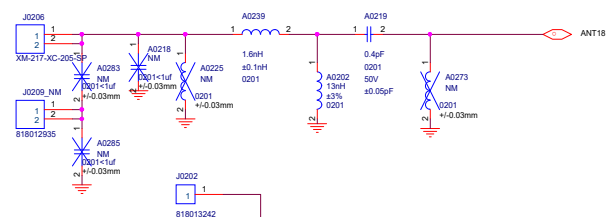
ANT19 GPS L5



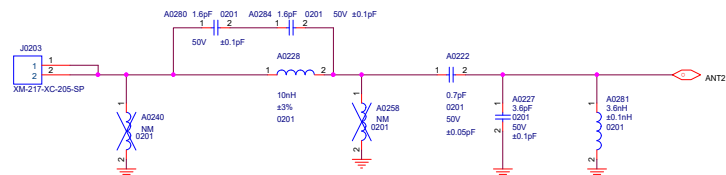
ANT17 WIFI 6E CHAIN0



ANT18 WIFI2.4G+WIFI6E CHAIN1



ANT2 LB(3)+B32



ANT0 CN
GLB MHB DM
MHB DM LB TRX
LB TRX N1/N3/N41 SA DM
n41 DRX N28 SA PRX
n1/n3/n7 DRX N41 NSA DRX

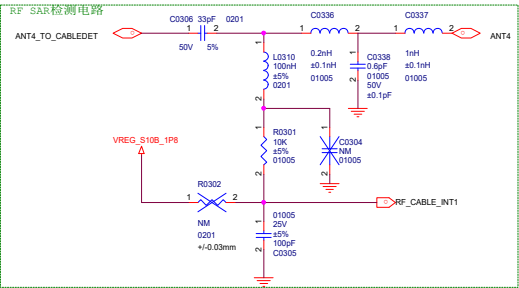
ANT1
GLB
LB DRX
CN
LB DRX
N28 SA DRX

ANT2
GLB
B20+N28 DRX
B32 PRX
CN
N5 PRX2
N8 PRX2

ANT3
GLB
MHB TRX
n1/n3/n7 PM
N41 PM

CN
MHB TRX
N1/N3/N41 SA TRX

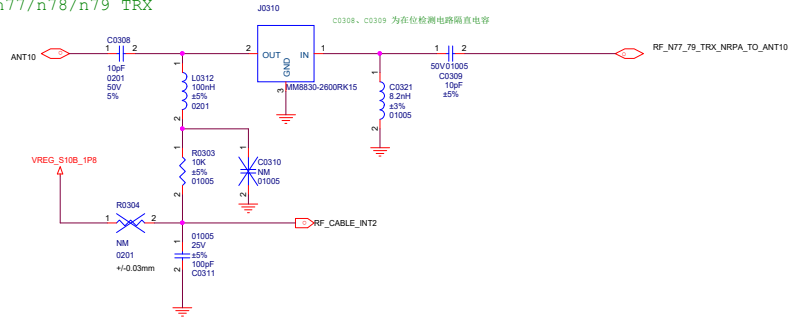
ANT4 CN
GLB MHB PM
MHB PM n41 NSA TRX
n41 TRX n1/n3/n41 SA PM
n1/n3/n7 TRX N41 SA TX1



ANT5
GLB CN
MHB DRX MHB DRX
n41 DM n41 DM
n1/n3/n7 DM n1/n3/n41 SA DRX
B32 DRX N41 NSA DM

ANT7(LDS)
GLB
B7 PRX IN ENDC B7_N41
CN
N41 NSA PM+

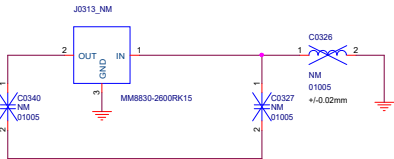
ANT10
n77/n78/n79 TRX



ANT11
N77/N78/N79 PM

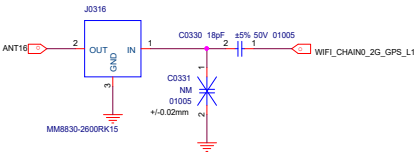
ANT12
N77/N78/N79 DM

ANT13 n77/n78/n79 DRX

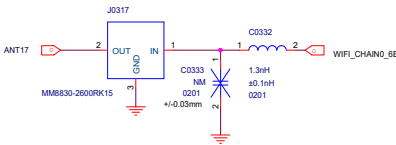


WIFI+GPS

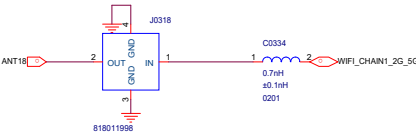
ANT16 GPS L1+WIFI2.4 CHAIN0



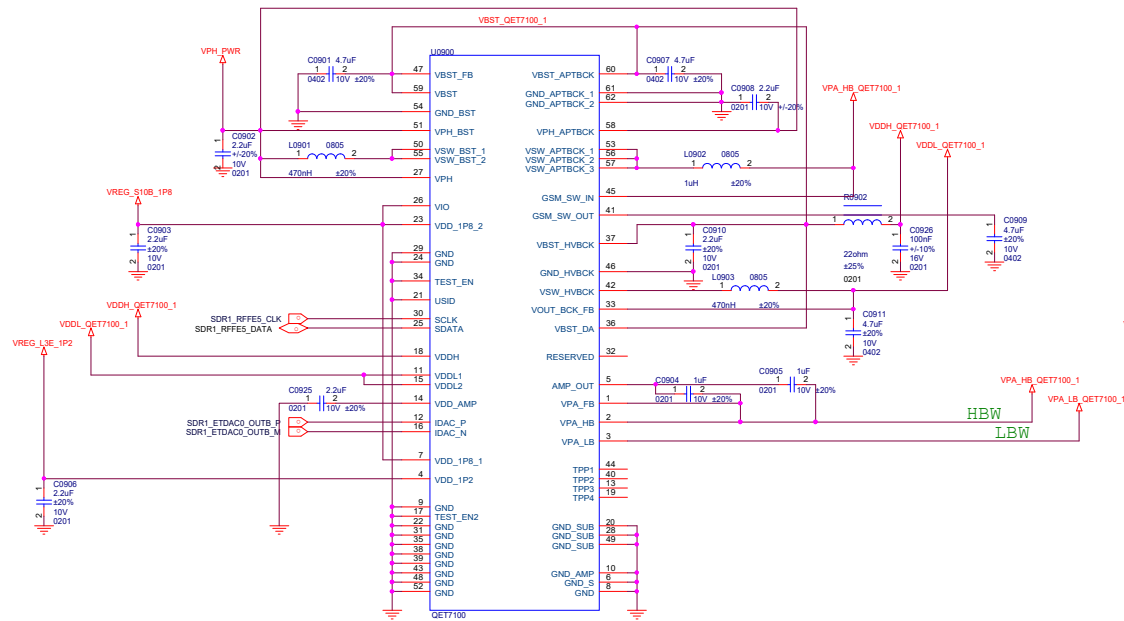
ANT17 WIFI 6E CHAIN0



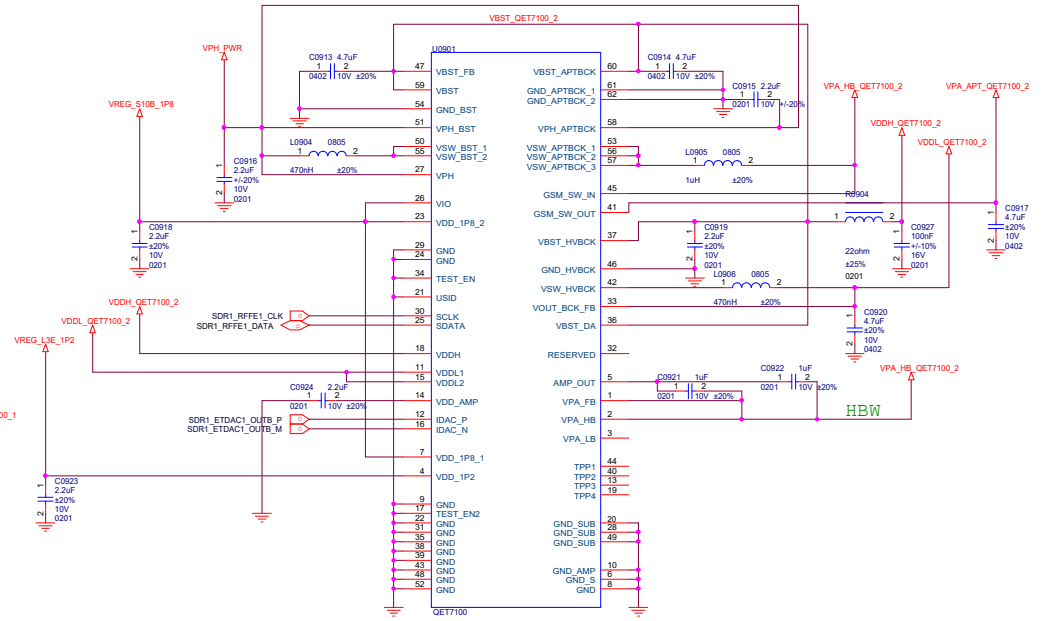
ANT18 WIFI2.4G+WIFI6E CHAIN1

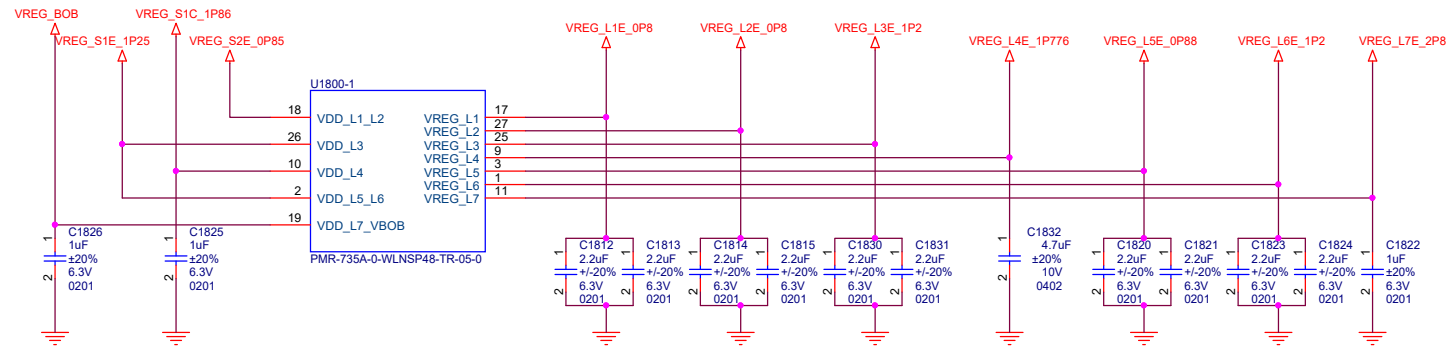
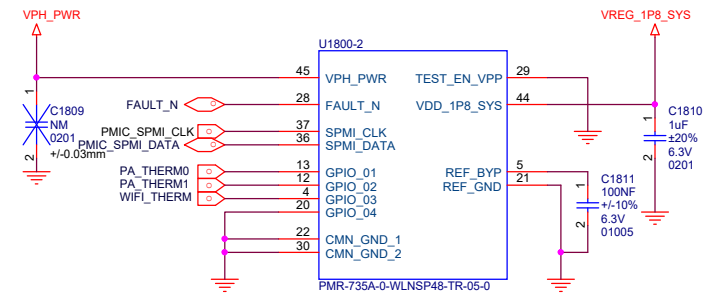
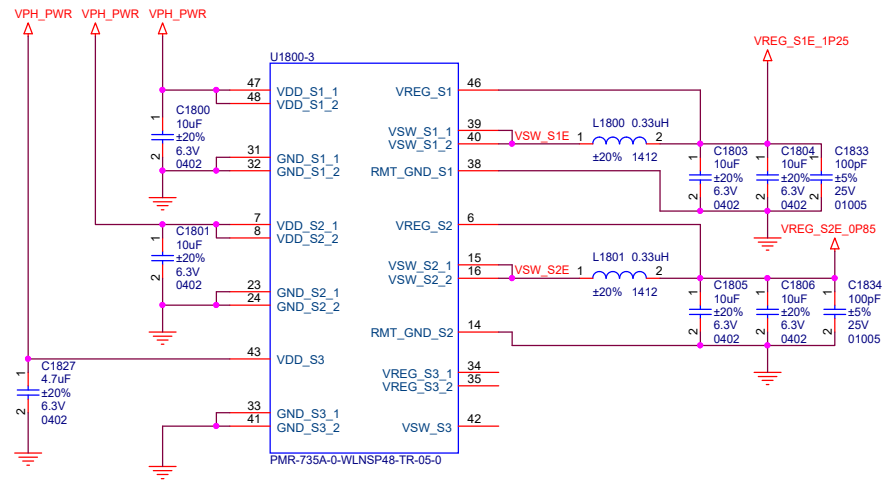


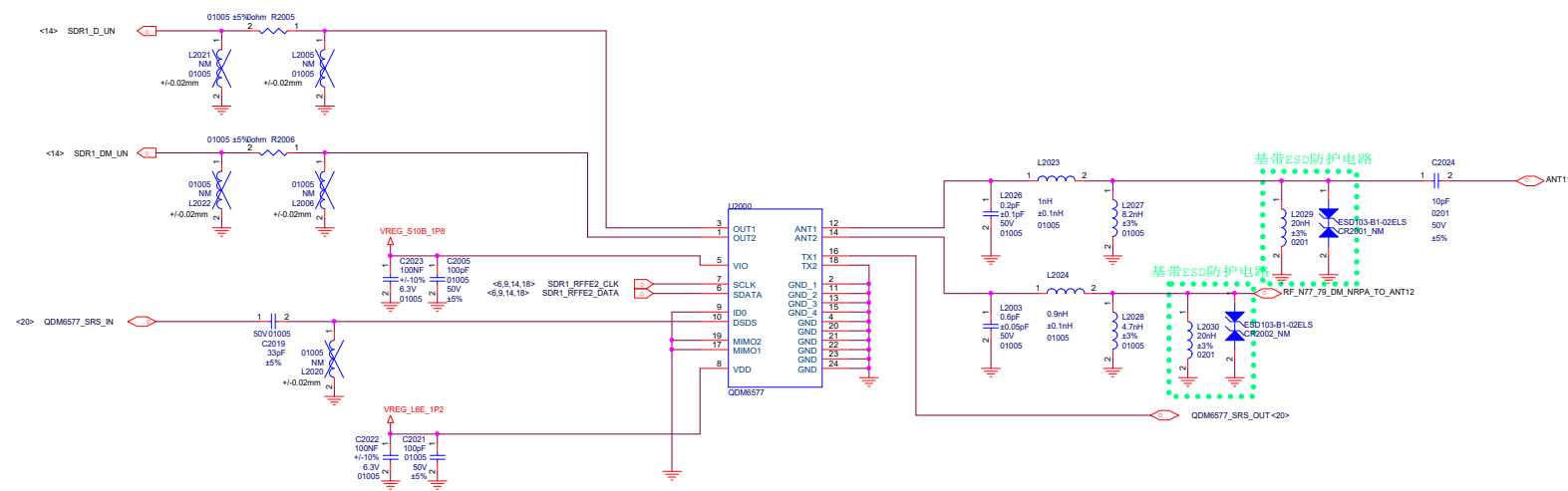
QET7100_1

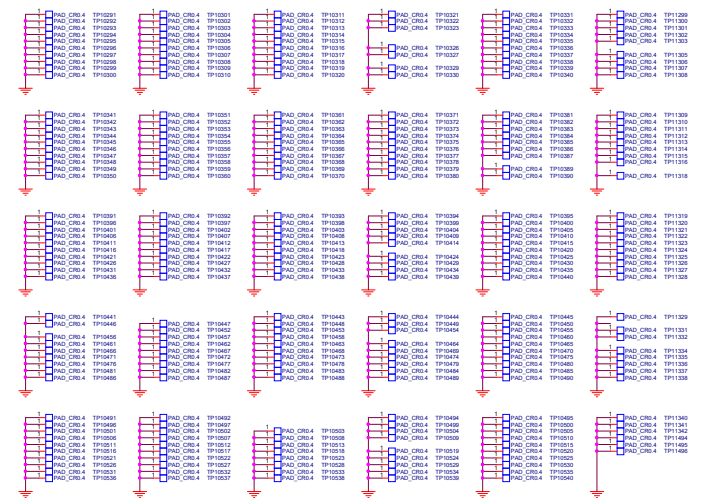


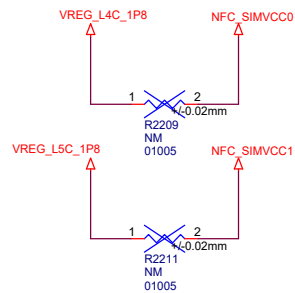
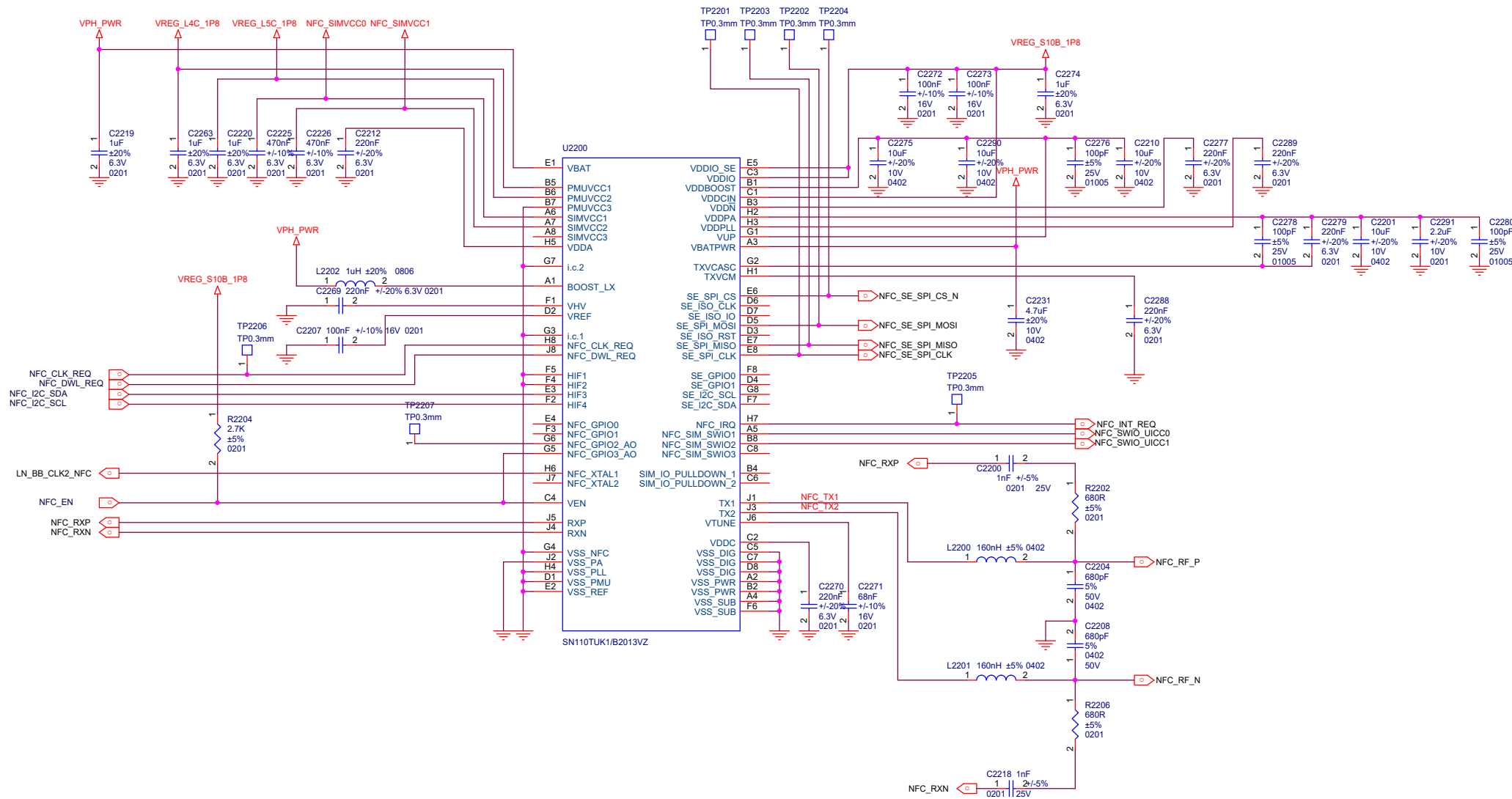
QET7100_2

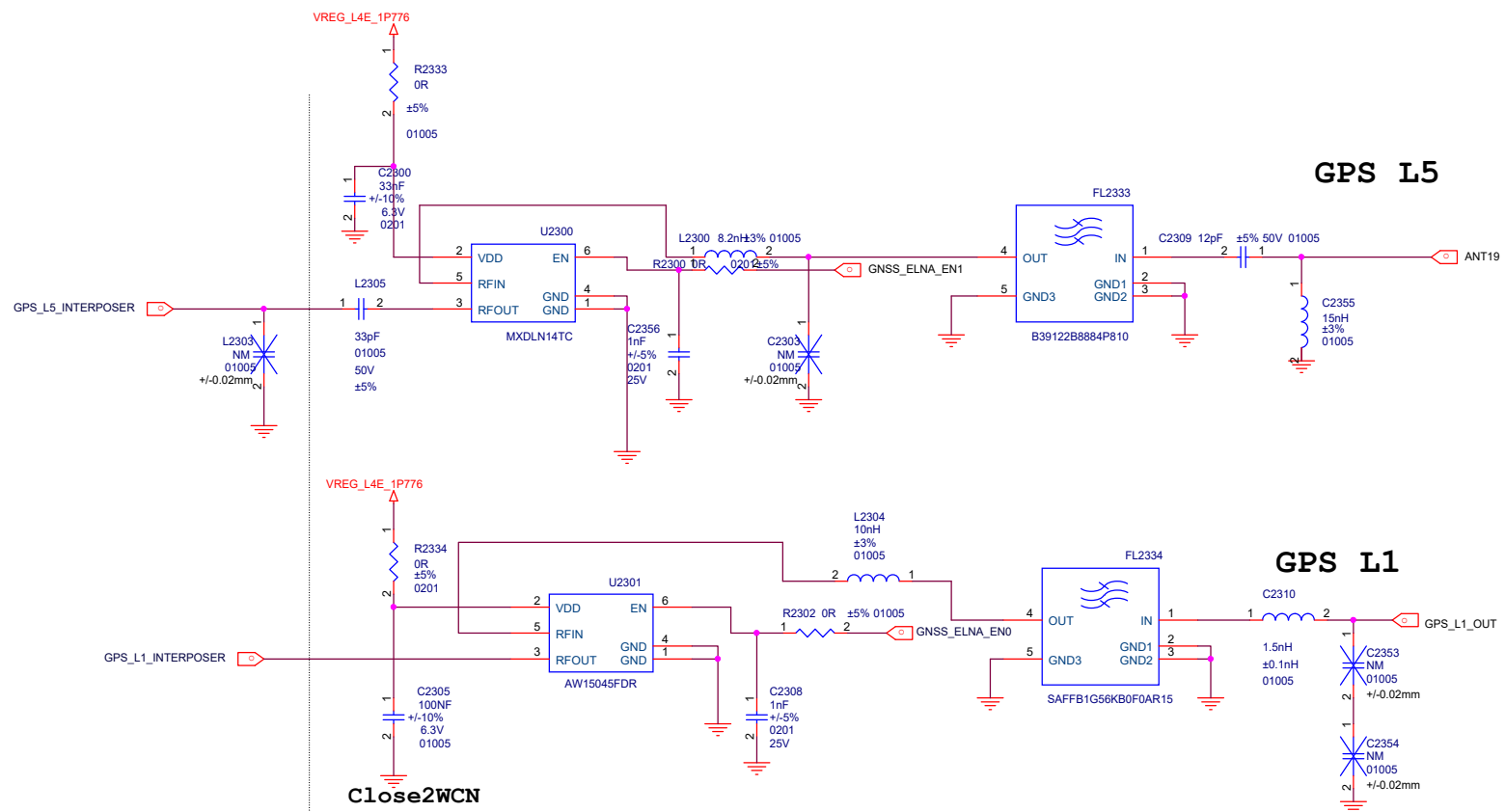


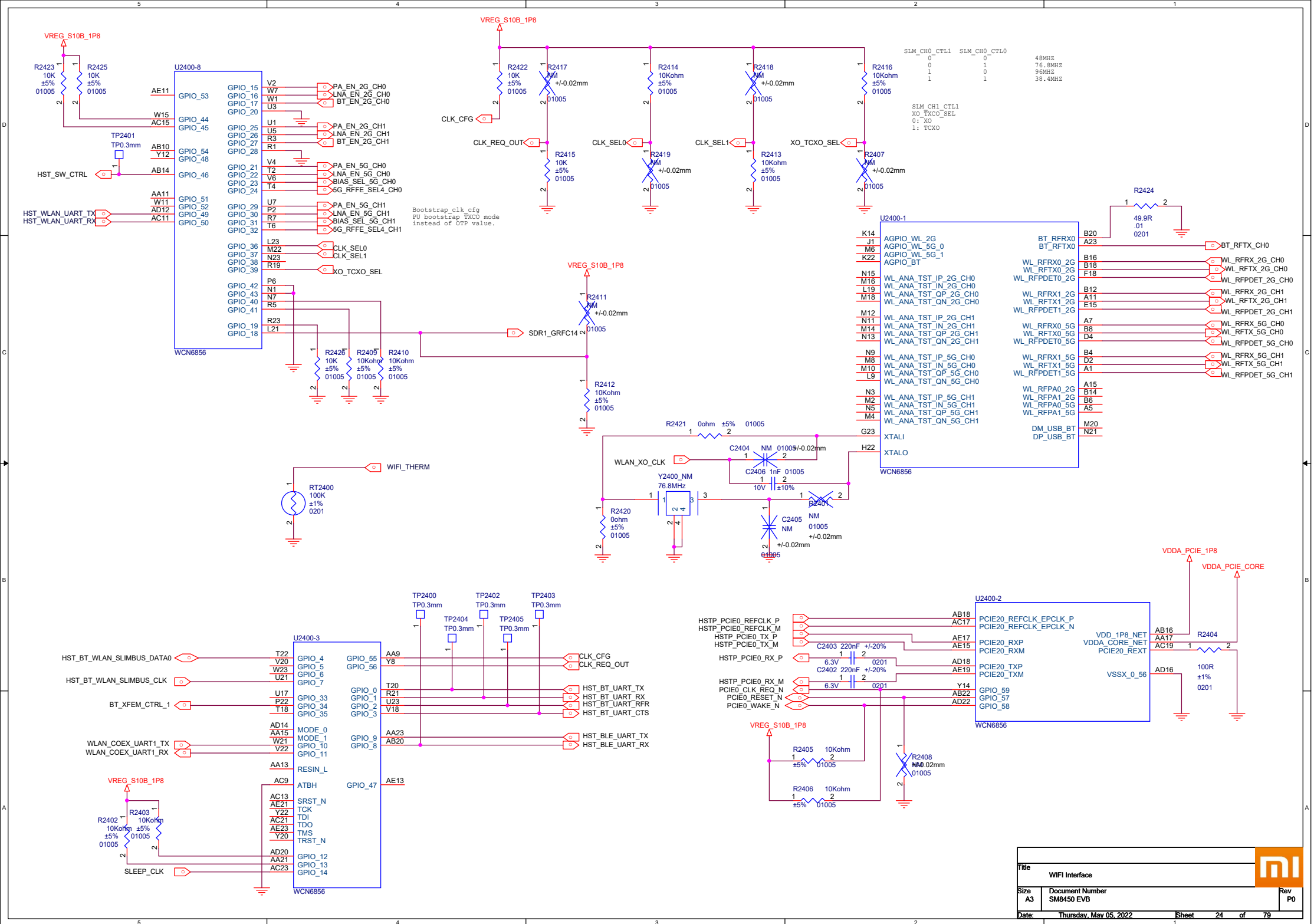




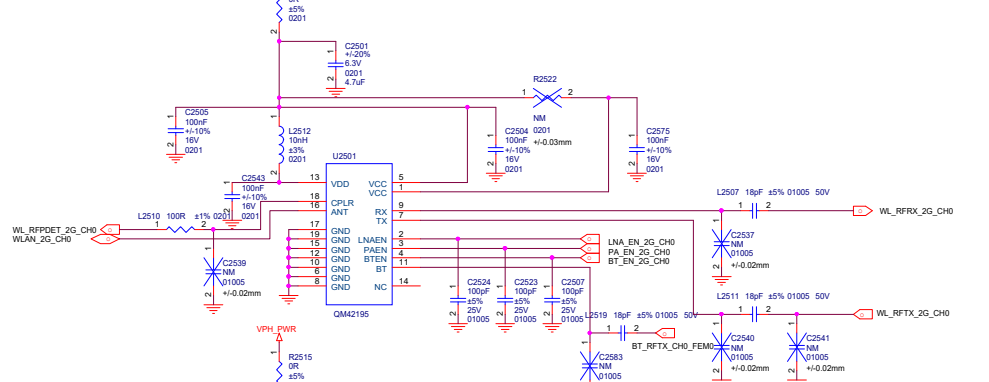




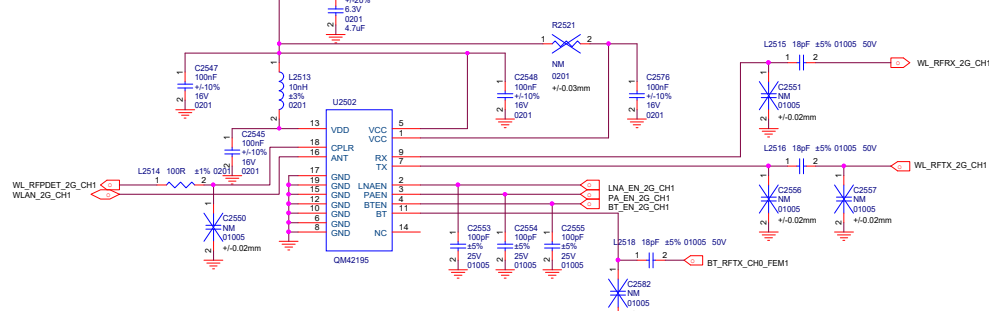




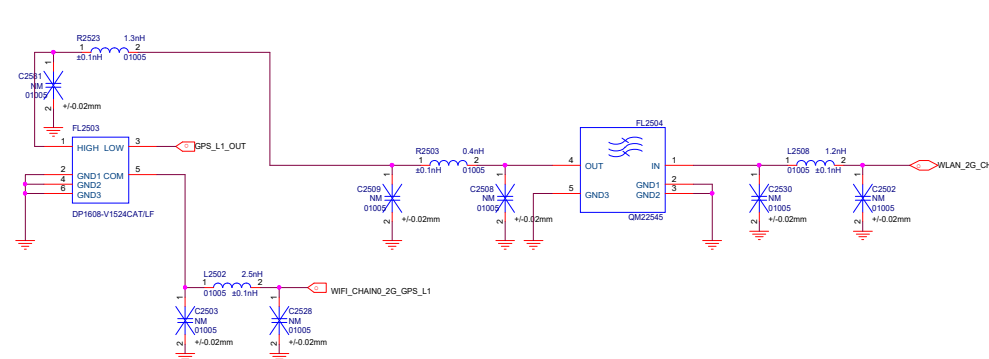
WIFI 2.4G CHAIN0



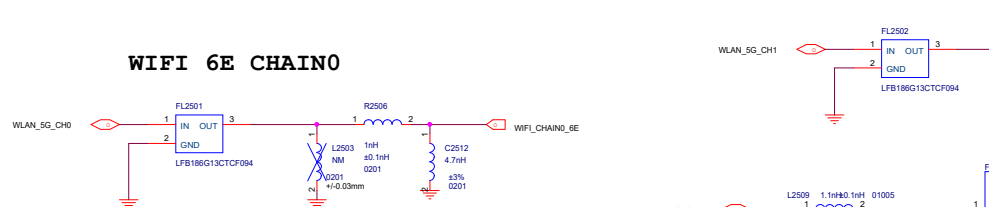
WIFI 2.4G CHAIN1



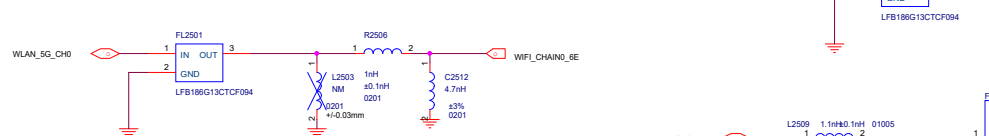
GPS L1+WIFI2.4 CHAIN0



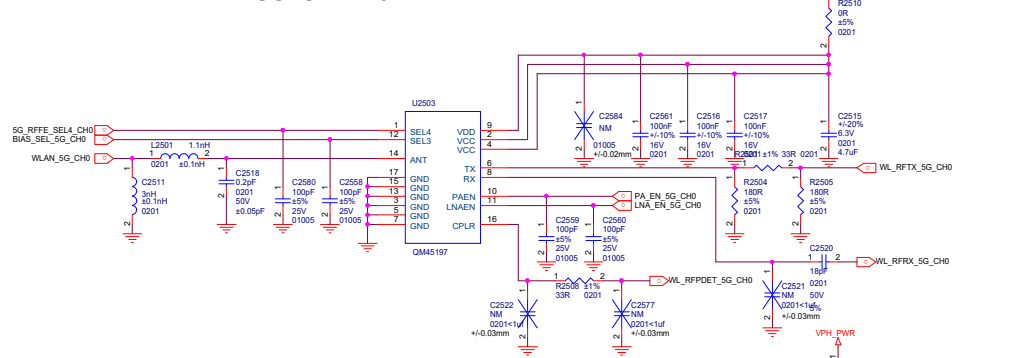
WIFI2.4G+WIFI6E CHAIN1



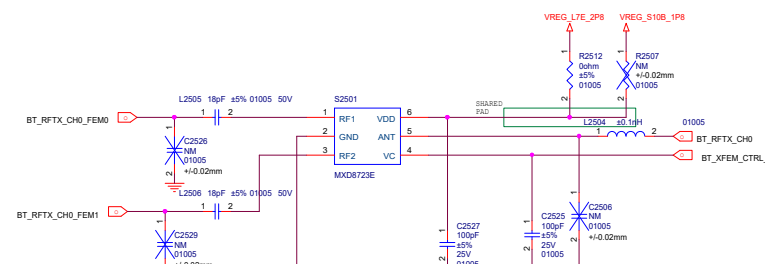
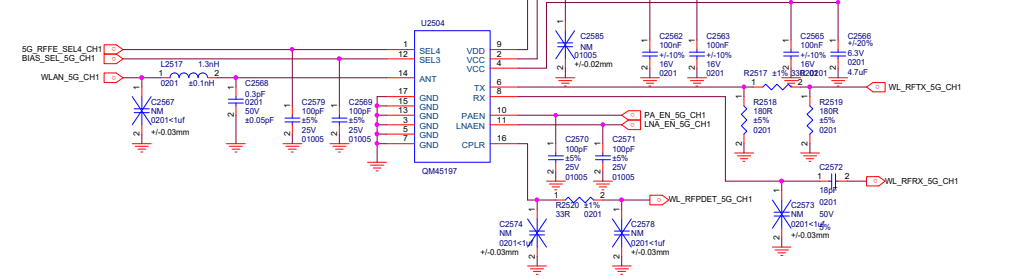
WIFI 6E CHAIN0



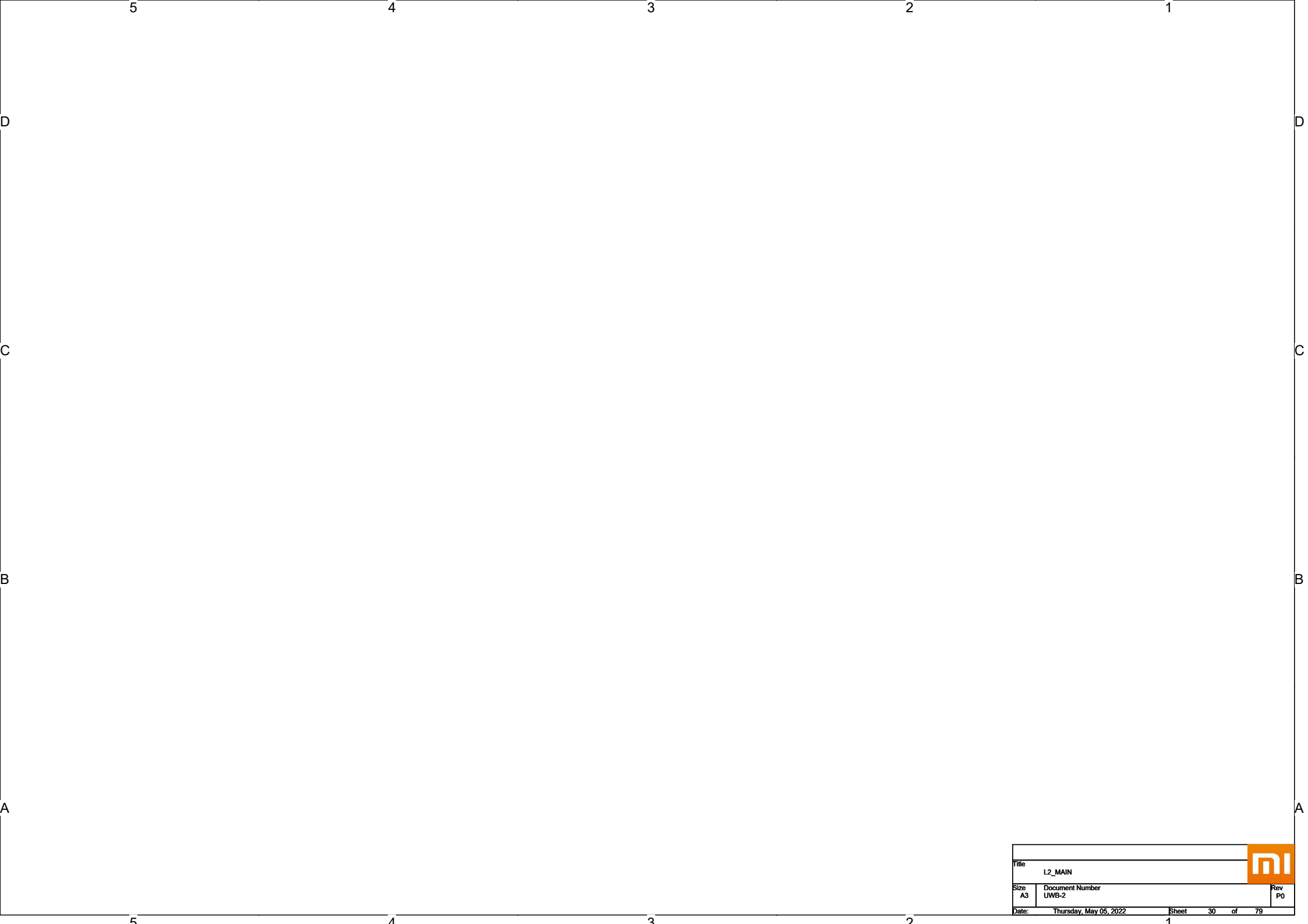
WIFI 5G CHAIN0



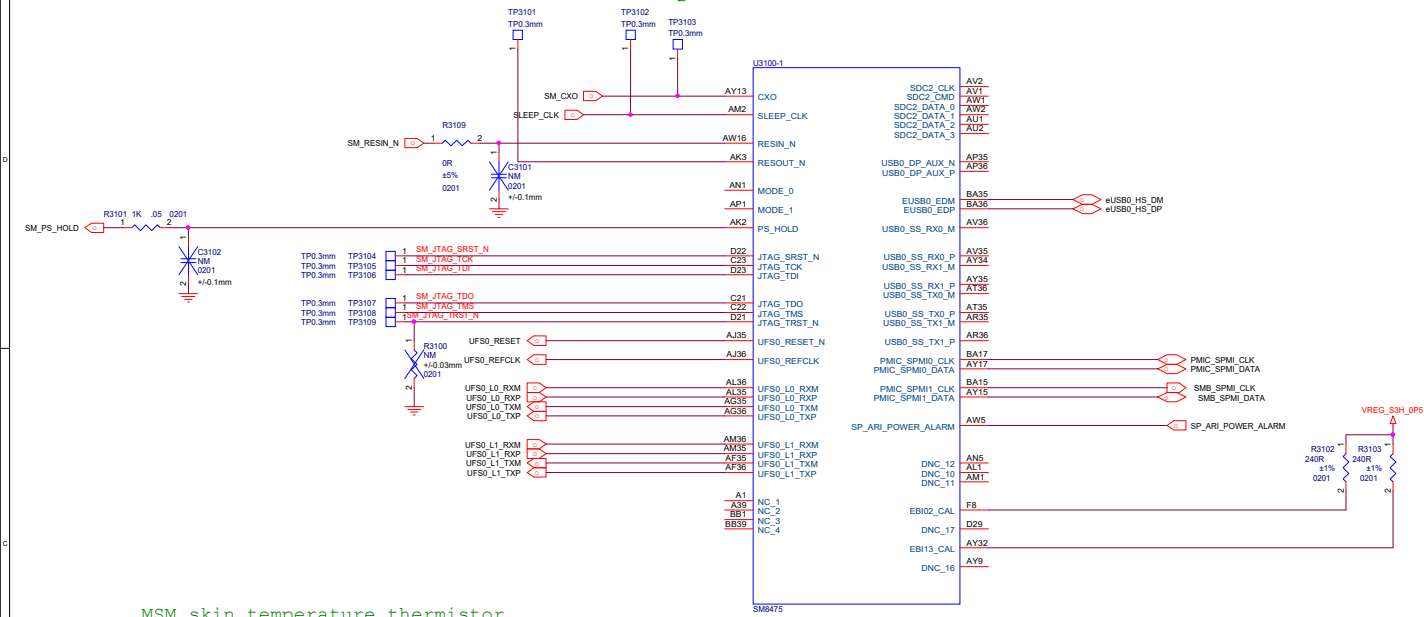
WIFI 5G CHAIN1



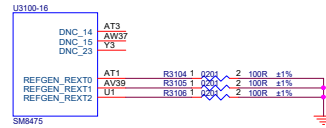
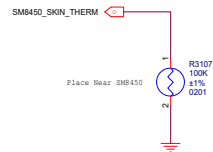
File	WIFI FEM	Rev	PO
Size	Document Number		
A2	SBM450 EVB		
Date:	Thursday, May 05, 2022	Sheet	25 of 79



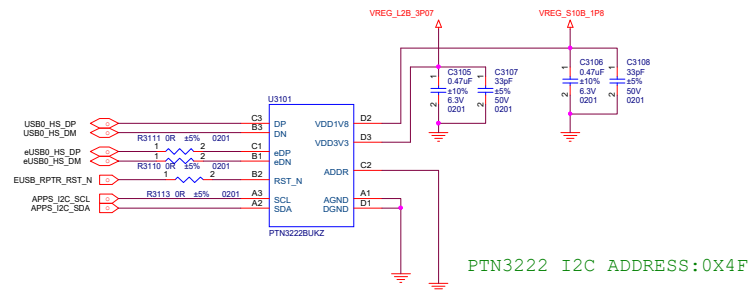
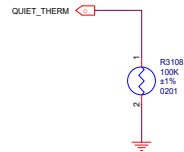
Place all ZQ resistors close to SM8350

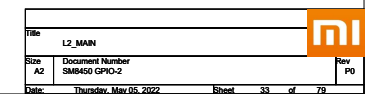


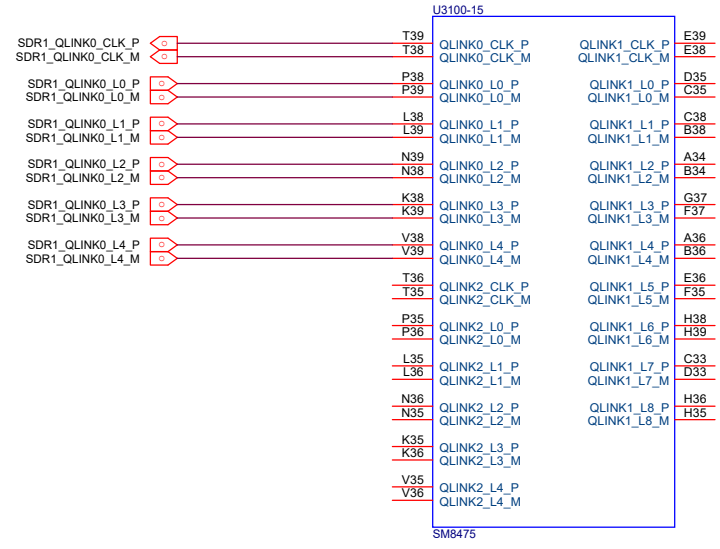
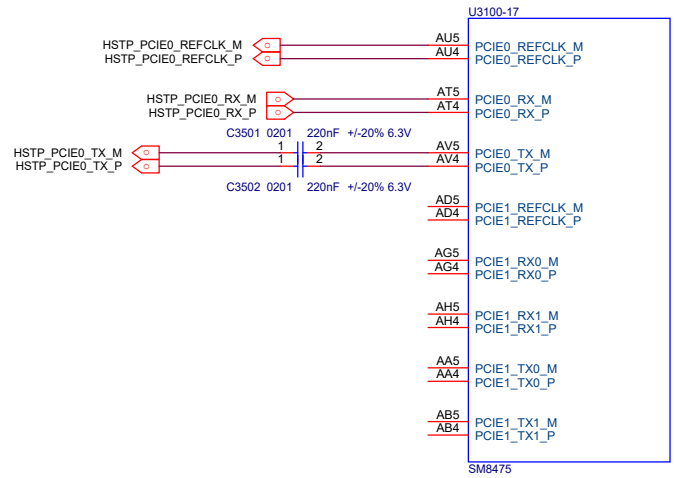
MSM skin temperature thermistor



Quiet thermistor Place on a cool place



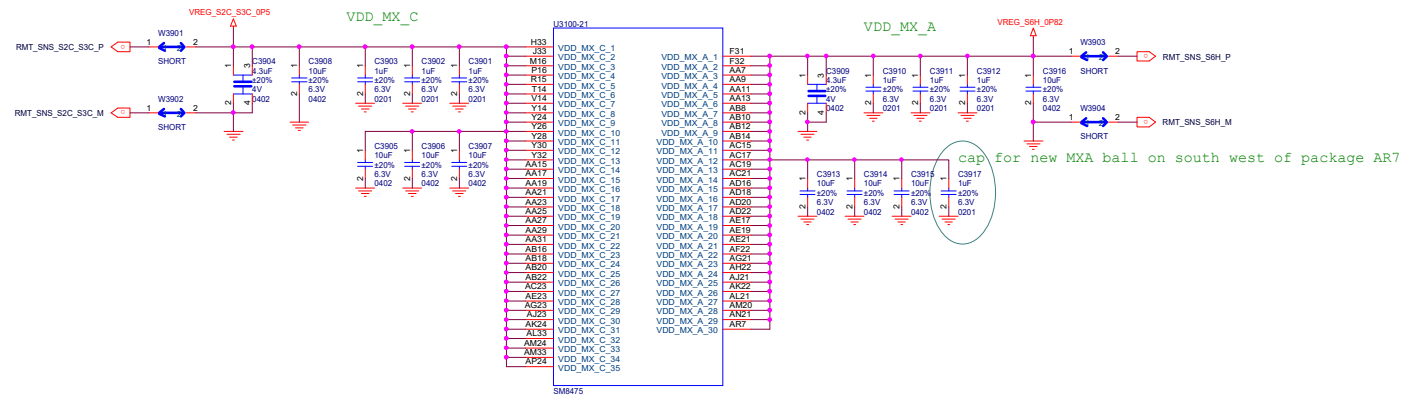


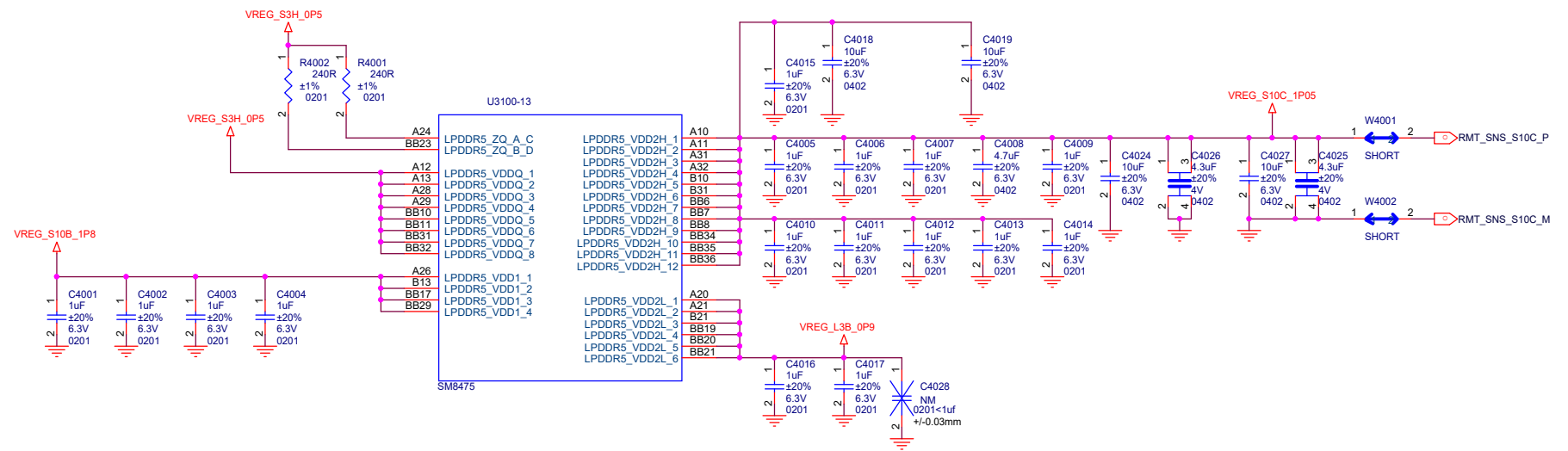


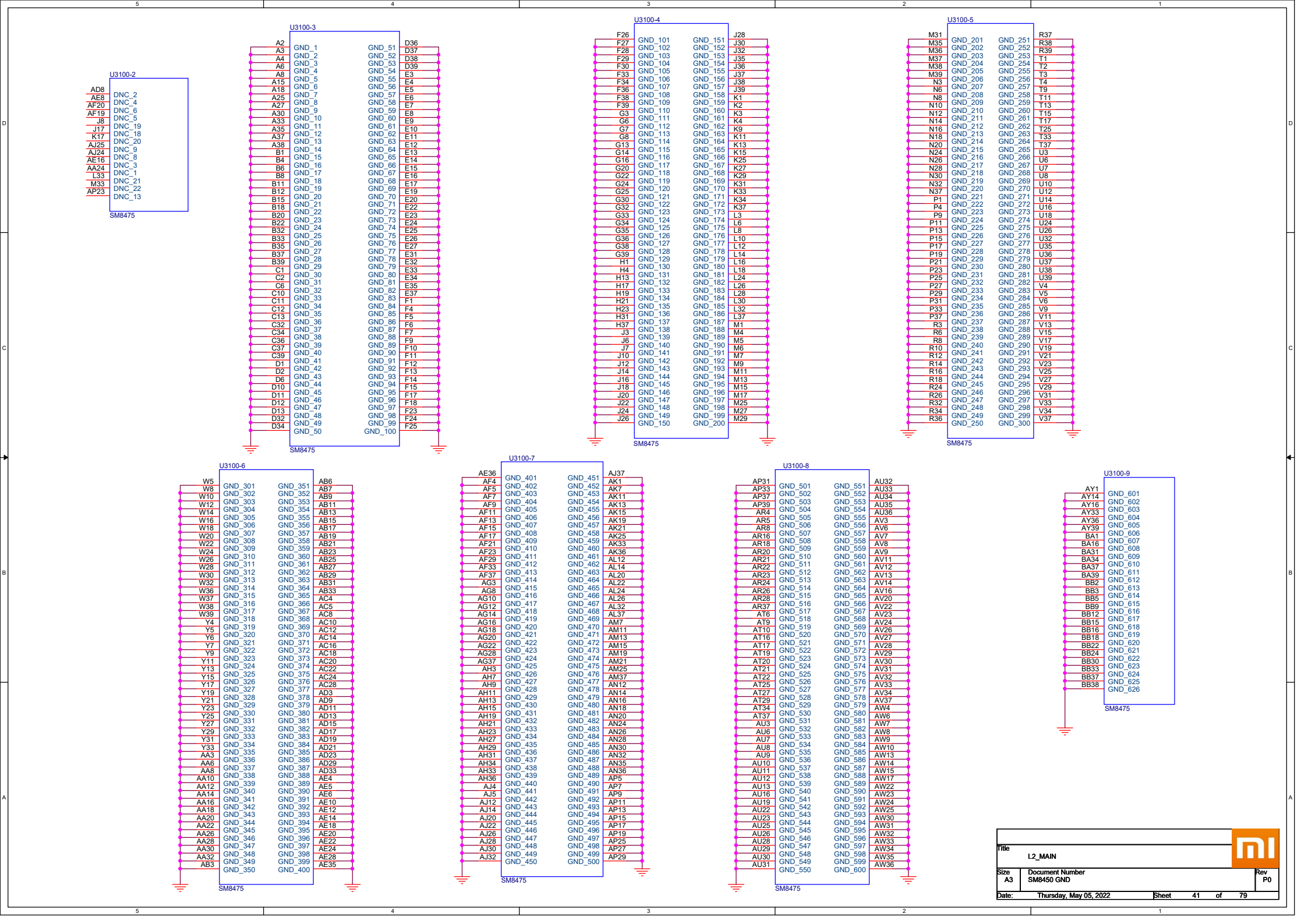
SDR1

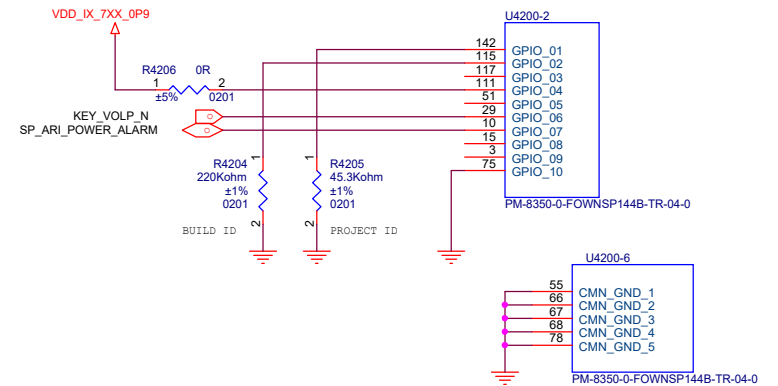
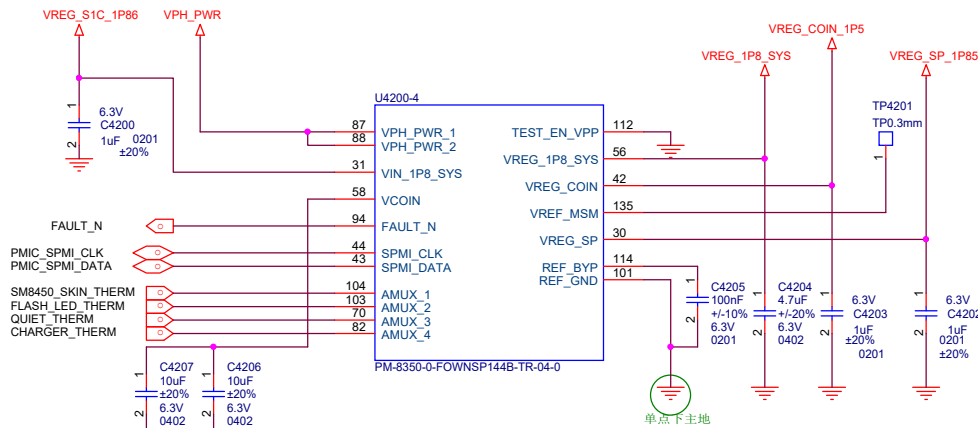
SDR2

QLINK1 for mmW



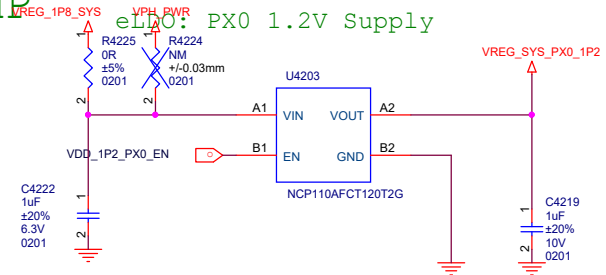
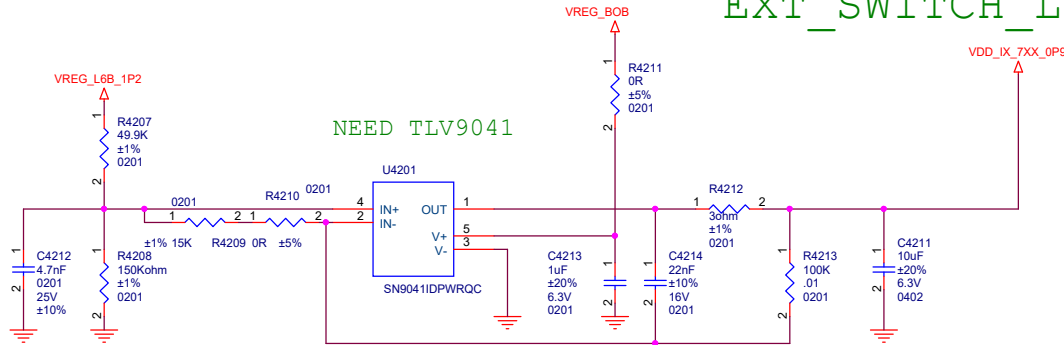






EXT_SWITCH_LDO_AMP

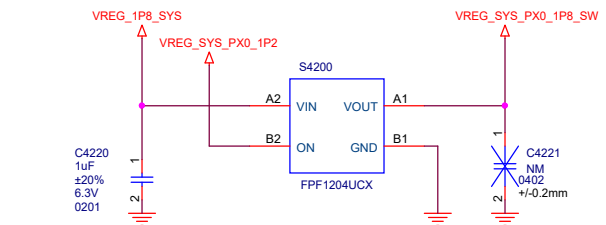
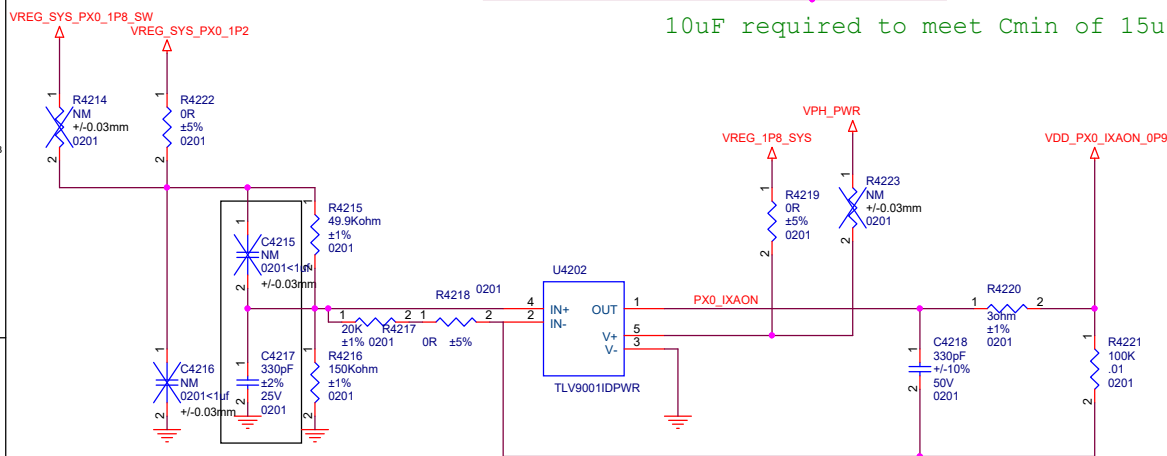
eLDO: PX0 1.2V Supply



NEED TLV9041

10uF required to meet Cmin of 15uF for the TLV9041

Load Switch: PX0 1.8V Supply



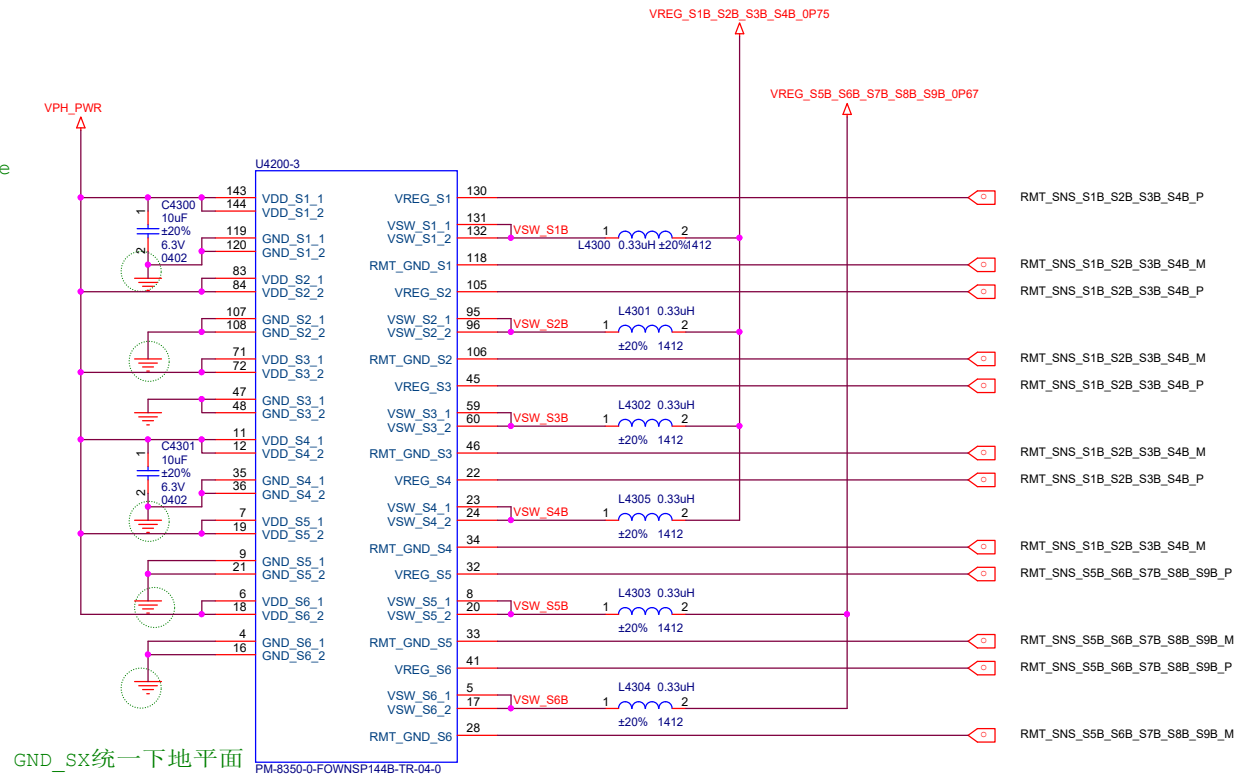
此处使用2% 精度

Input Cap shared between load switch and eLDO
eLDO和load SW布局不在一起时，input 增加1uF电容

LDO	Irate (ma)	Default On Voltage(V)	Working Voltage Min	Working Voltage Normal	Working Voltage Max	Net Name	Capless	Cout(uf)
L1H	1200	0.824	0.75	0.75	0.83	LPI_MX/NAV	NO	4.7~22
L2H	1200	0.912	0.87	0.91	0.93	BB QLink0/1/2,PCle1	NO	4.7~22
L3H	400	0.912	0.87	0.91	0.96	Caster QL 9L	NO	2.2~10
L4H	300	1.808	1.504	TBD	2.0	Not used	YES	1~10

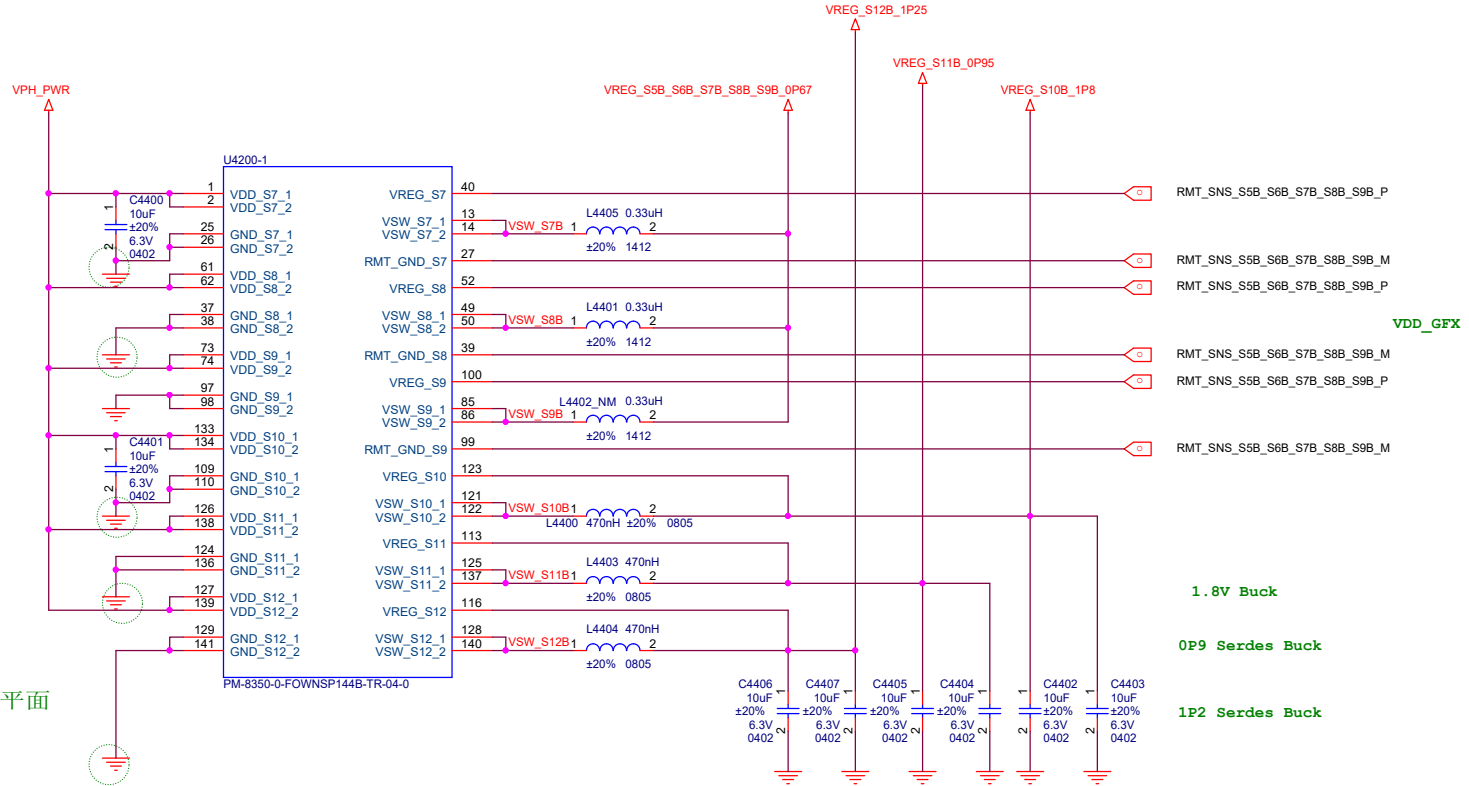
					
Title					
L2_MAIN					
Size	Document Number				Rev
A3	PM8350 XO&CTRL				
Date:	Thursday, May 05, 2022			Sheet	42 of 79

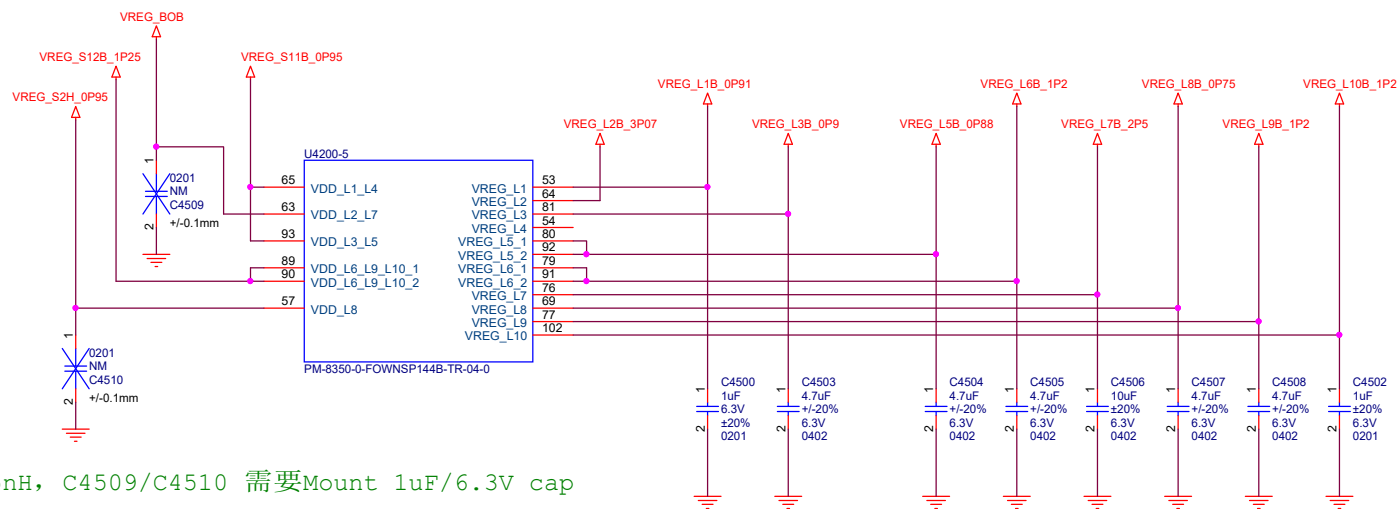
VDD_SX统一接VPH_PWR shape



VDD_SX统一接VPH_PWR shape

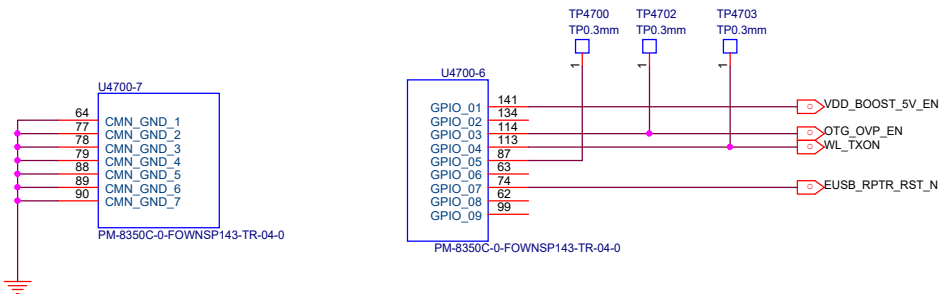
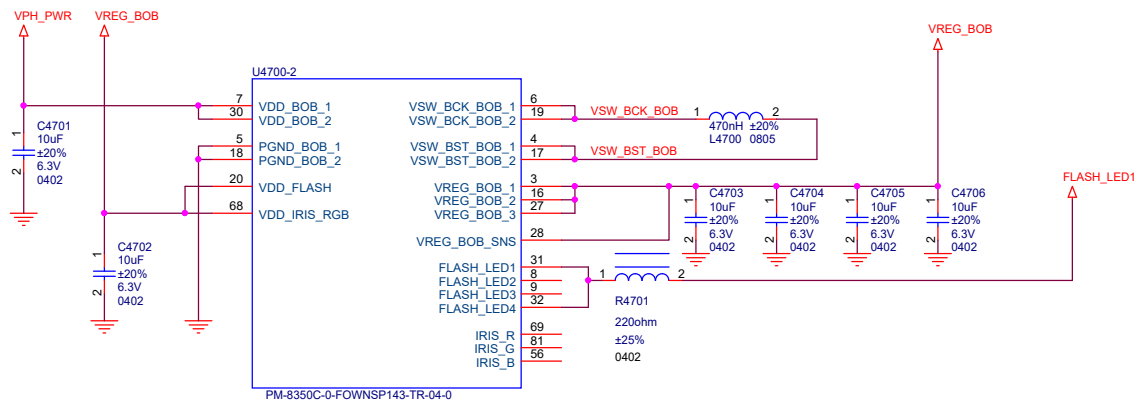
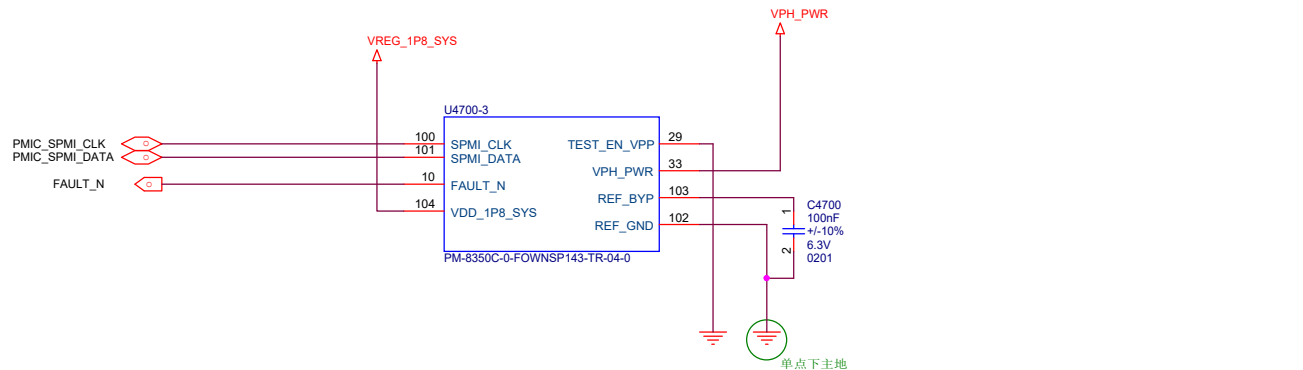
GND_SX统一一下地平面





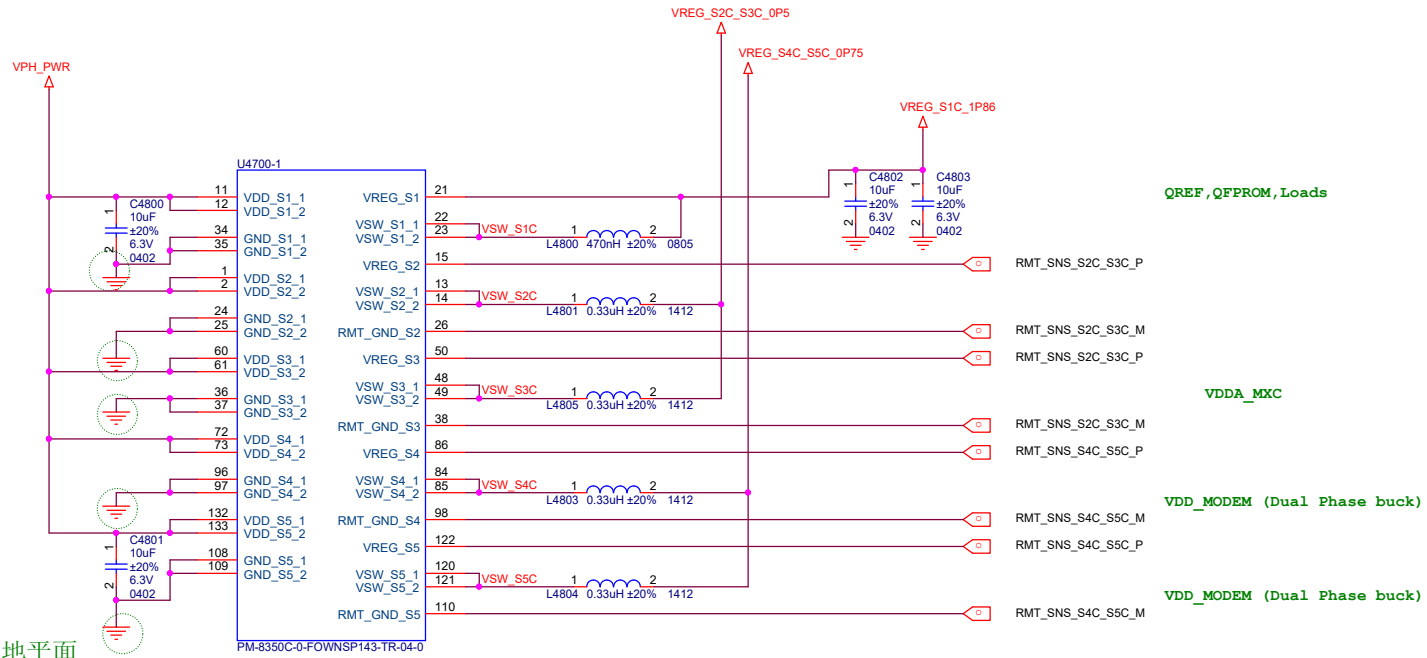
输入ESL大于15nH, C4509/C4510 需要Mount 1uF/6.3V cap

LDO	Irate (ma)	Default On Voltage(V)	Working Voltage			Net Name	Capless	Cout(uf)
			Min	Normal	Max			
L1B	300	0.912	0.82	0.88	0.92	USB_SS_DP	NO	1~10
L2B	150	3.072	2.97	3.072	3.3	USB PHY, USB_HS_3P1	YES	0.47~5
L3B	1200	0.904	0.56	0.904	0.97	LPDDR5_VDD2L	NO	4.7~22
L4B	300	--	0.32	TBD	1.304	Not used	NO	1~10
L5B	1200	0.88	TBD	TBD	TBD	SM8450 analog	NO	4.7~22
L6B	1200	1.2	TBD	TBD	TBD	SM8450 VDD analog	NO	4.7~22
L7B	600	2.504	TBD	TBD	TBD	UFS0_VCC	YES	4.7~23.5
L8B	1200	0.824	TBD	0.752	TBD	LPI_CX	NO	4.7~22
L9B	1200	1.2	TBD	TBD	TBD	PX10, UFS_VCCQ	NO	4.7~22
L10B	300	1.2	TBD	TBD	TBD	PMK8450 VDD_Active	NO	1~10



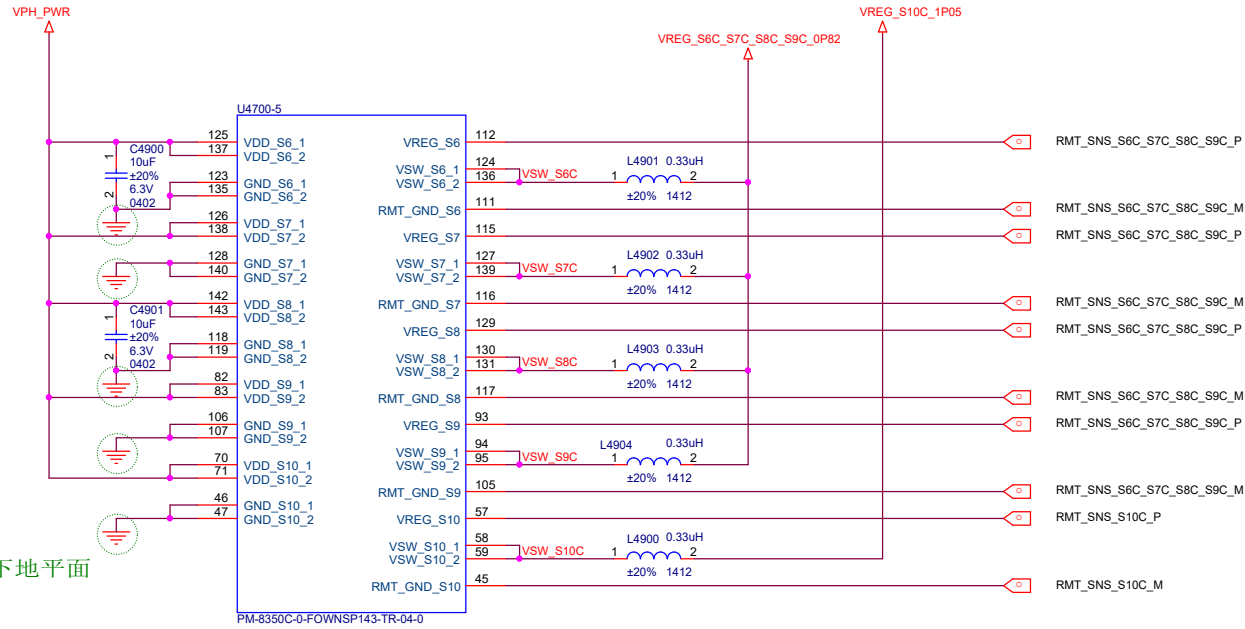
VDD_SX统一接VPH_PWR shape

GND_SX统一一下地平面



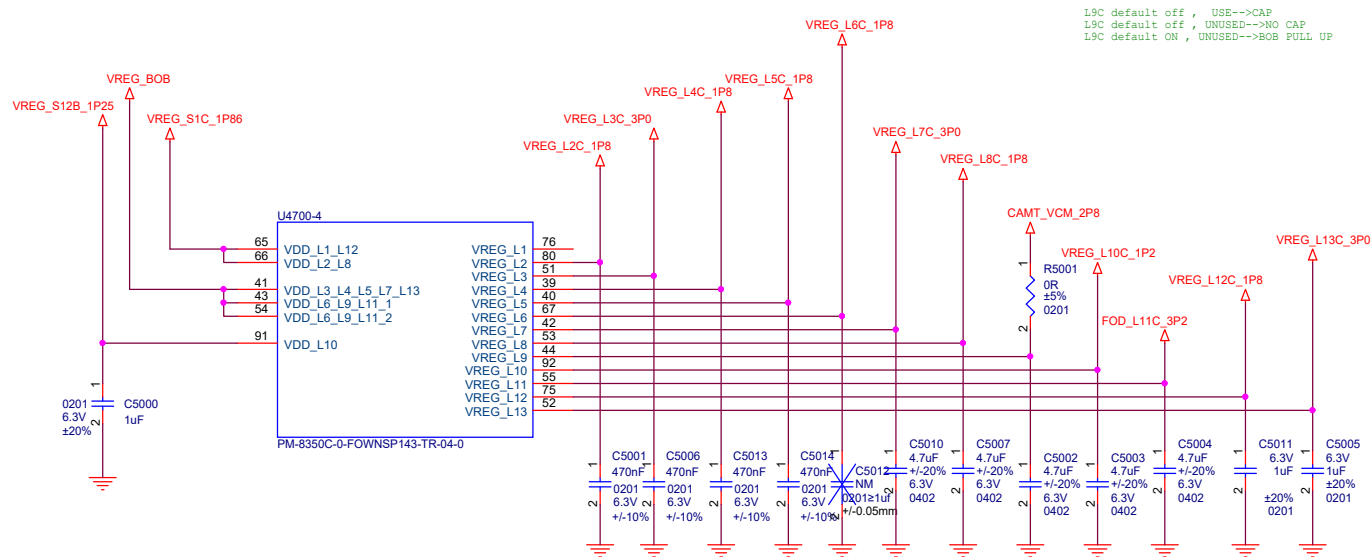
VDD_SX统一接VPH_PWR shape

GND_SX统一下地平面

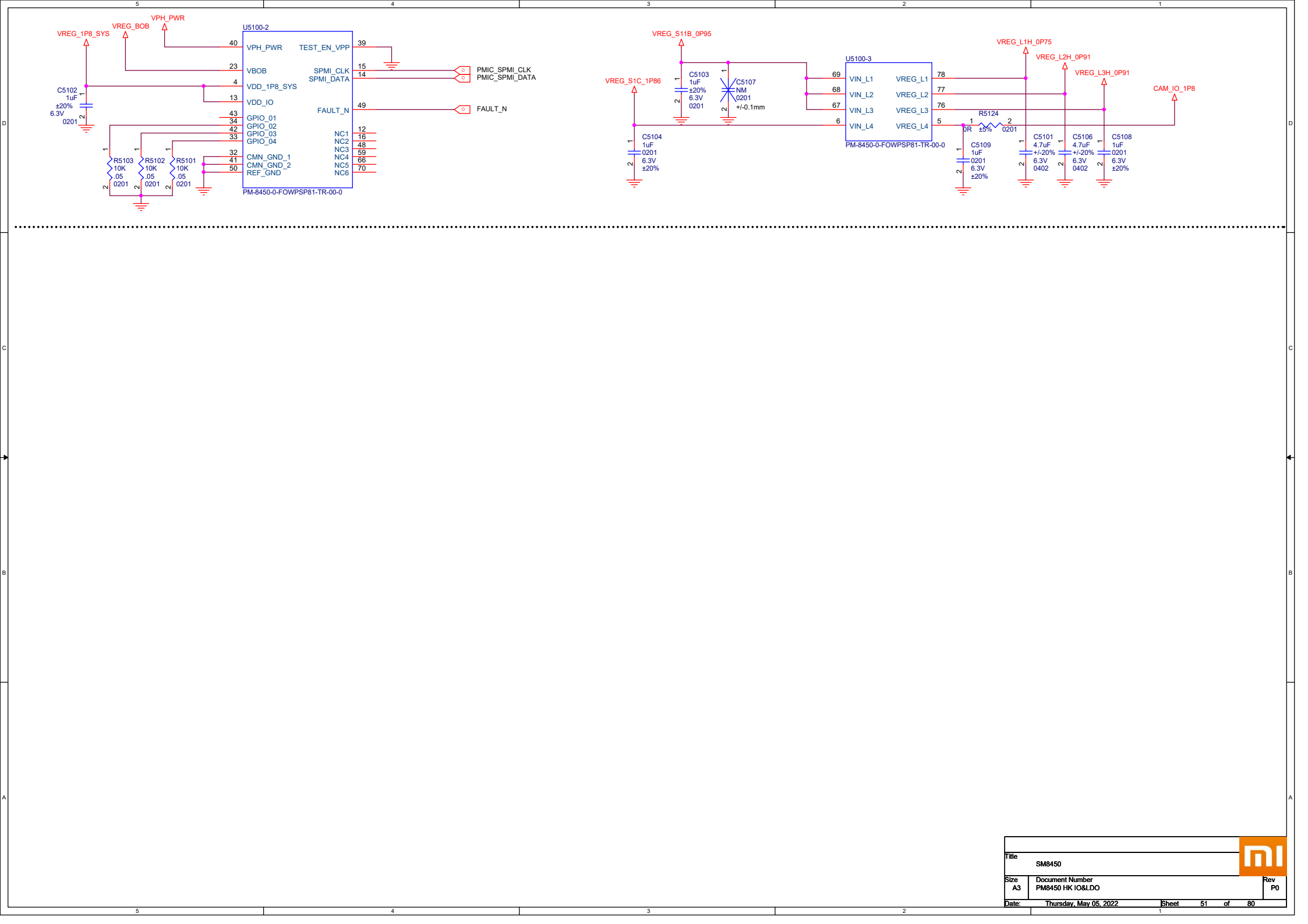


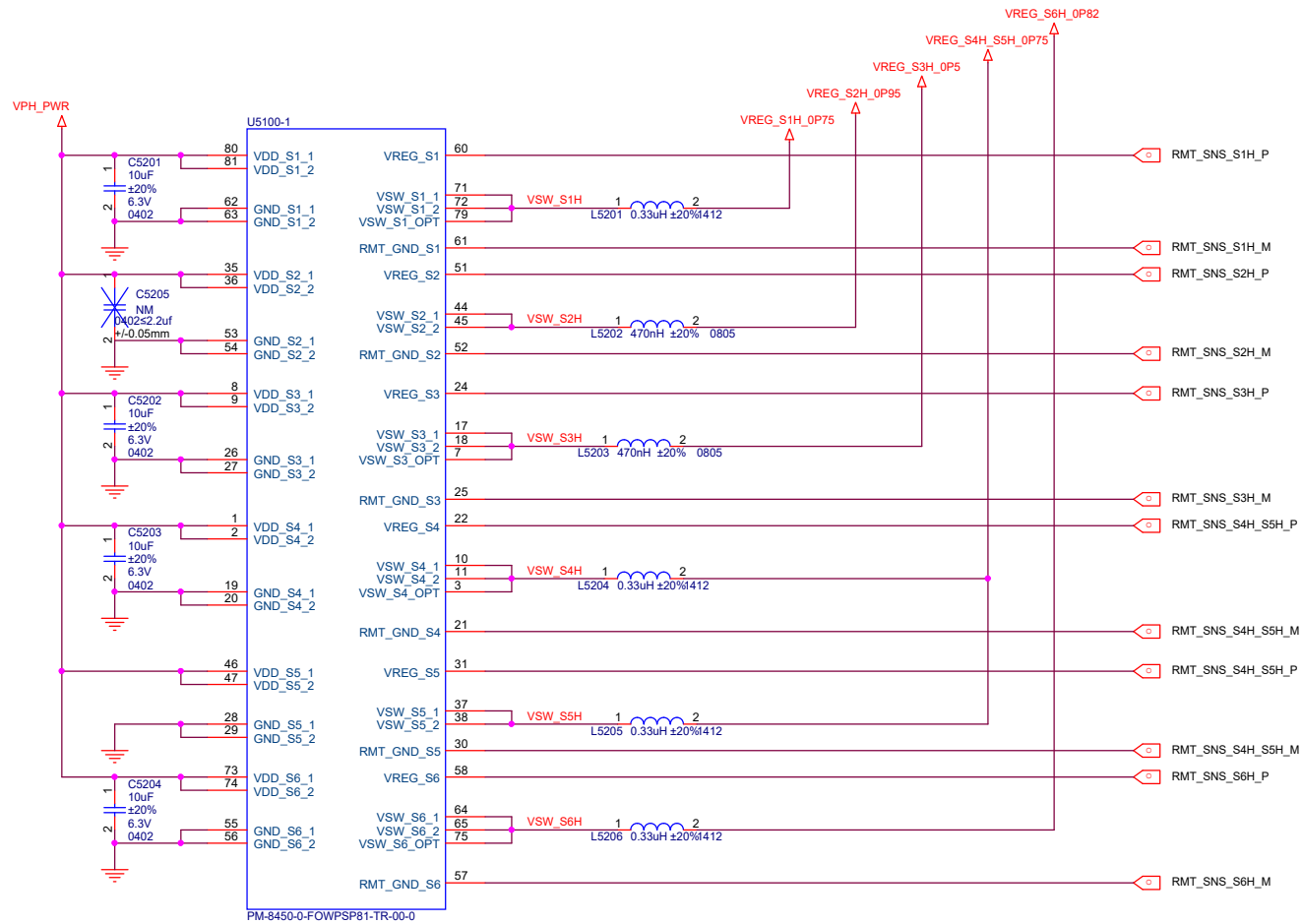
VDD_CX

S10C, LPDDR5 VDD2H/LPDDR4 VDD2



LDO	Irate (ma)	Default On Voltage(V)	Working Voltage			Net Name	Capless	Cout(uf)
			Min	Normal	Max			
L1C	150	1.8	1.6	1.8	1.98	USB2_1P8	YES	1~5
L2C	150	-- --	1.6	1.8	1.98	TP_VDDIO	YES	1~5
L3C	150	-- --	2.7	3.296	3.6	Sar Sensor	YES	0.47~5
L4C	150	-- --	1.65	1.808	3.3	SIM1	YES	0.47~5
L5C	150	-- --	1.65	1.808	3.3	SIM2	YES	0.47~5
L6C	150	1.8	1.76	1.8	1.84	PX2(兼容)	YES	0.47~5
L7C	600	-- --	3.0	3.008	3.6	TP 3.0	YES	4.7~23.5
L8C	600	-- --	1.71	1.8	1.95	Sensor	YES	4.7~23.5
L9C	600	2.96	2.7	2.96	3.6	SD Card(Not used)	YES	4.7~23.5
L10C	1200	-- --	0.99	1.2	1.25	OLED_VDD1P2	NO	4.7~22
L11C	600	-- --	TBD	2.504	TBD	FOD	YES	4.7~23.5
L12C	150	-- --	1.65	1.8	1.95	OLED_VDD1P8	YES	1~5
L13C	150	-- --	2.65	3.0	3.544	OLED_VCC3P0	YES	0.47~5





VDD APC0

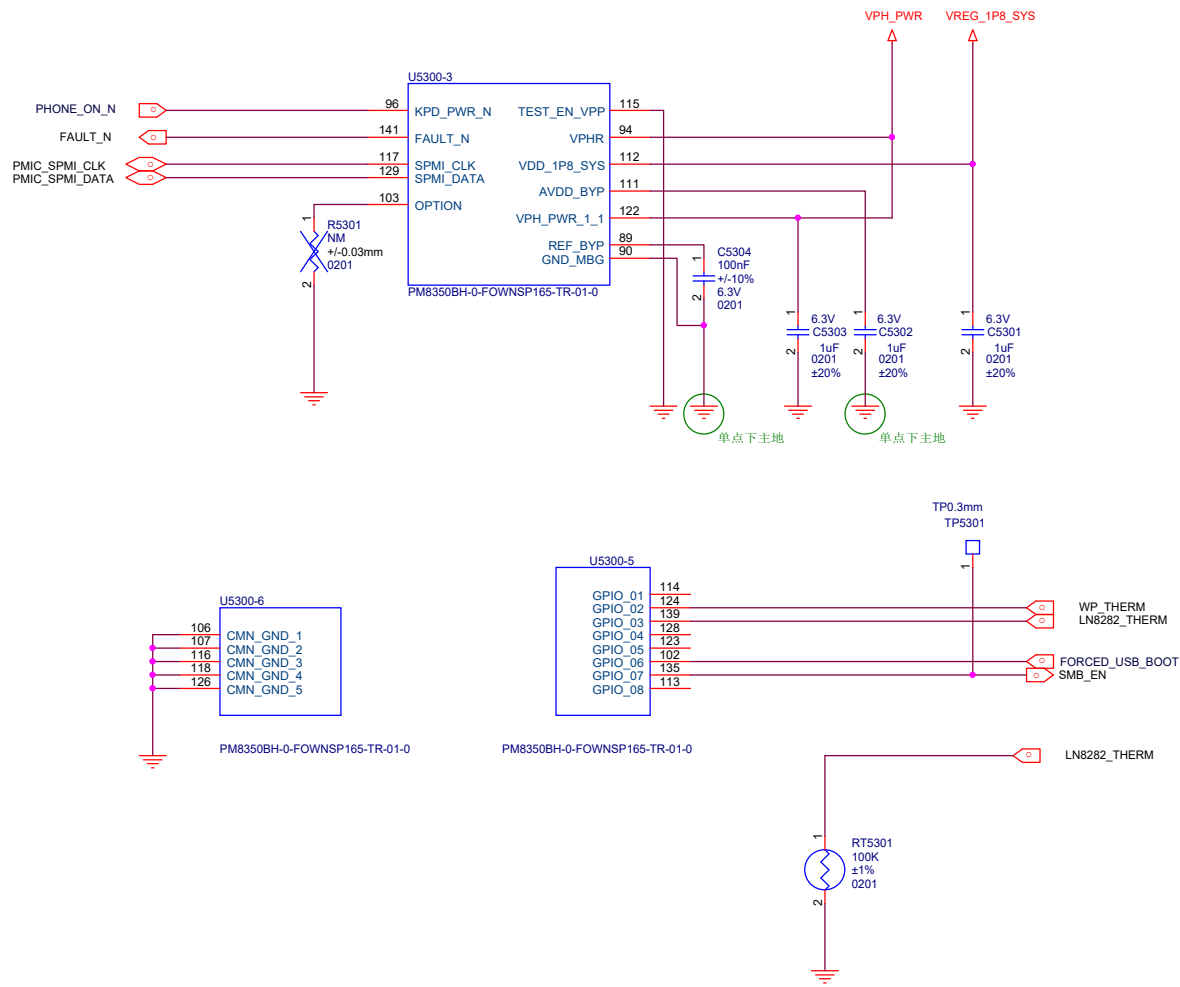
LPDDR5 PHY

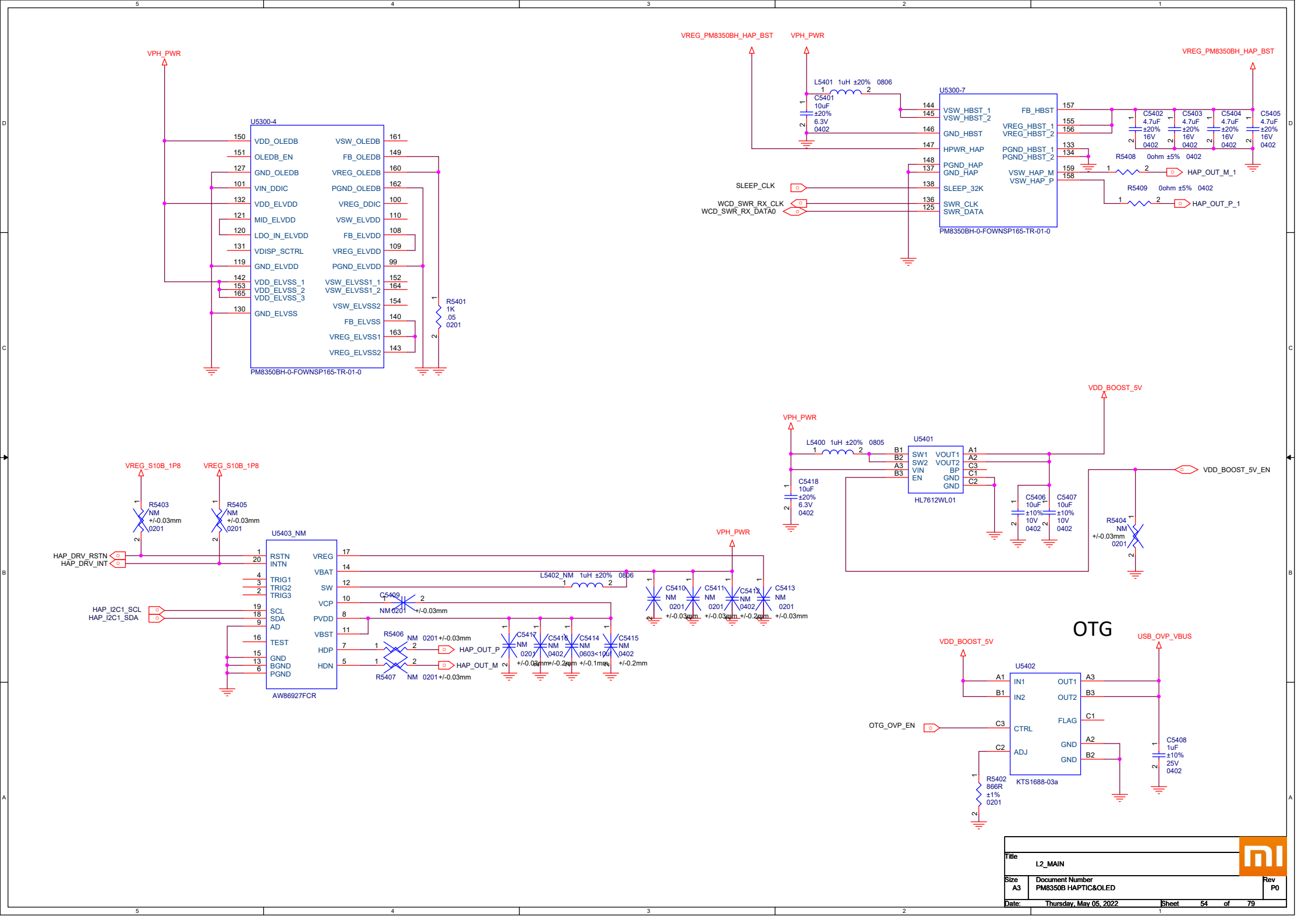
LPDDR5 VDDQ

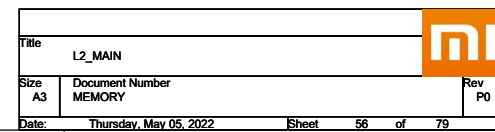
VDD MM

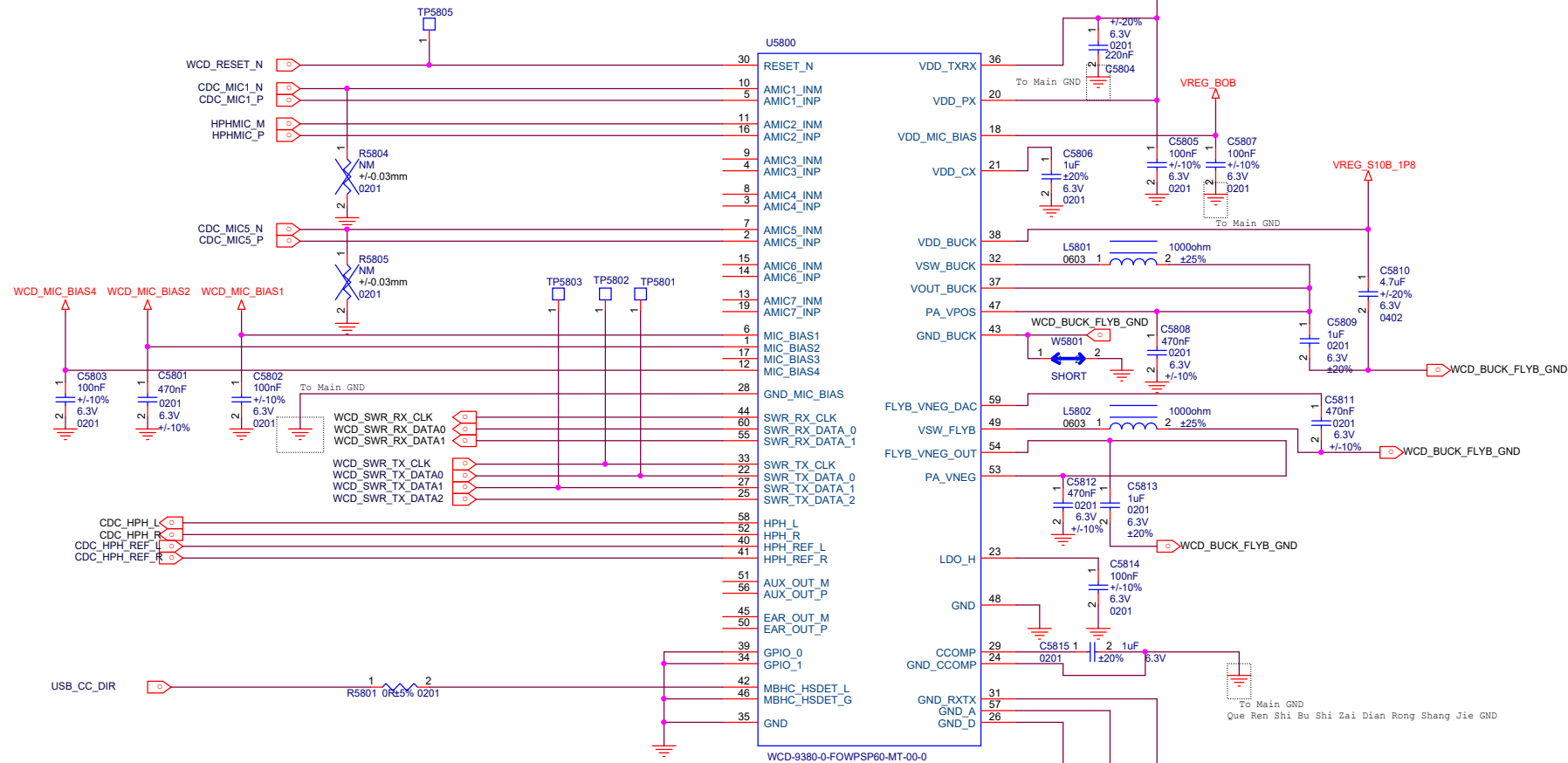
VDD MX A





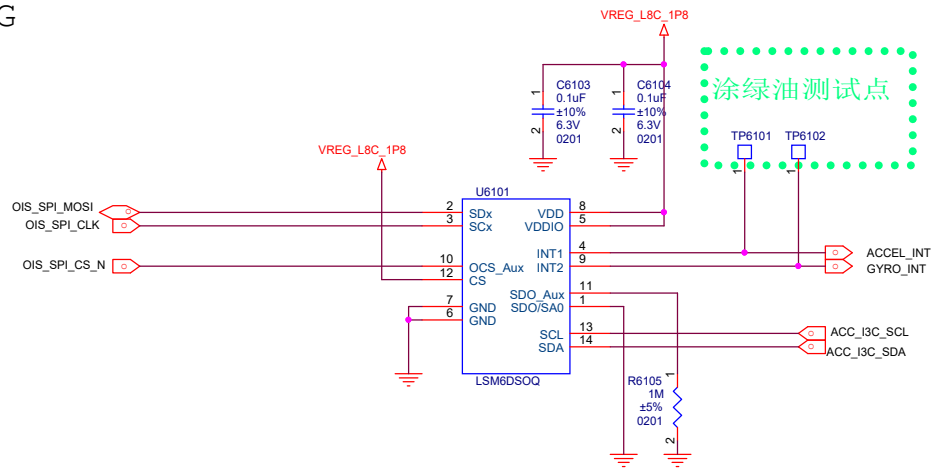




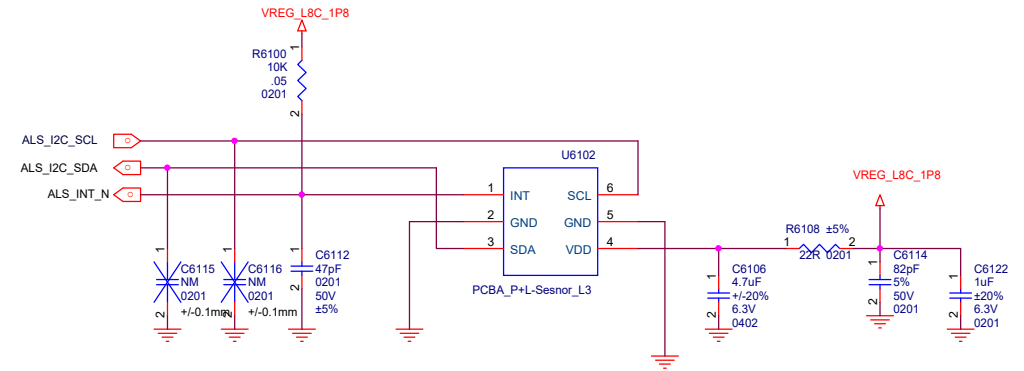


NOTE 1: ROUTE AN ISOLATED GROUND TRACE FROM CCOMP_REF TO THE CAP GROUND, THEN VIA TO THE MAIN GROUND PLANE
 NOTE 2: PLACE VDD_BUCK CAP AS CLOSE TO CHIP EDGE AS POSSIBLE AND MINIMIZE LOOP INDUCTANCE
 NOTE 3: USE SHORT AND WIDE TRACES FOR BUCK AND FLYBACK RELATED NETS
 NOTE 4: ANALOG MIC INPUTS SHOULD BE ISOLATED WITH GROUND FROM NOISY OR SWITCHING SIGNALS AND SUPPLIES

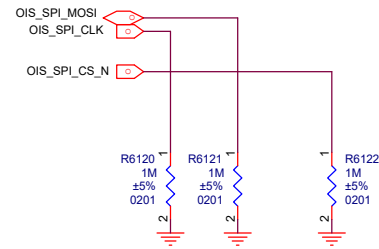
A+G



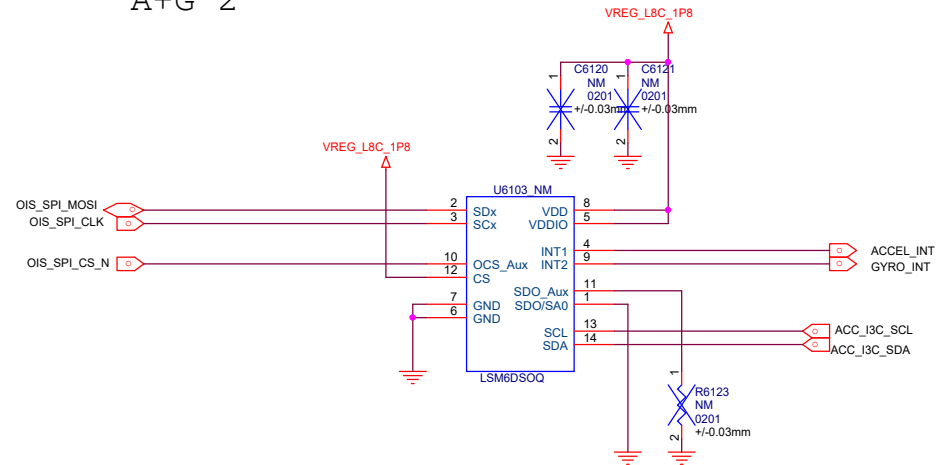
P_L-sensor



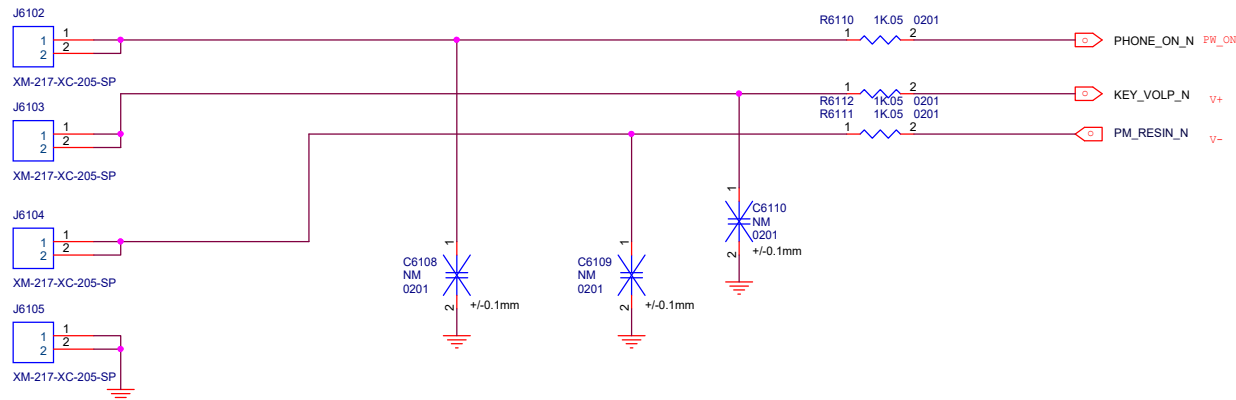
PRESSURE



A+G 2

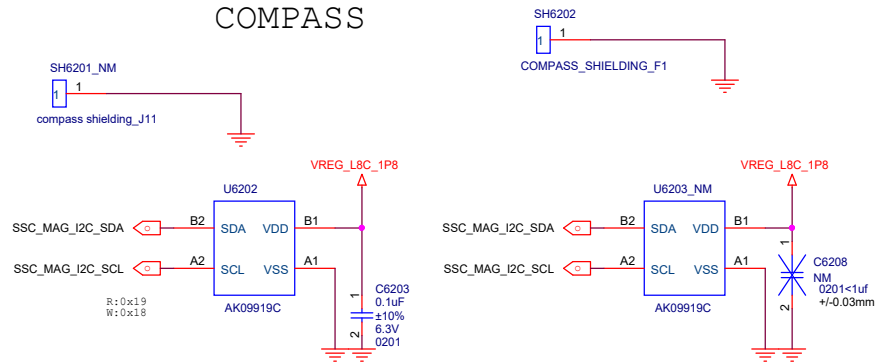


Sidekey

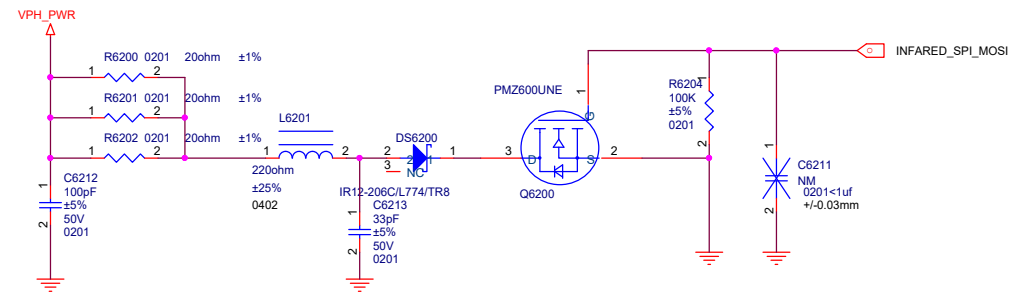


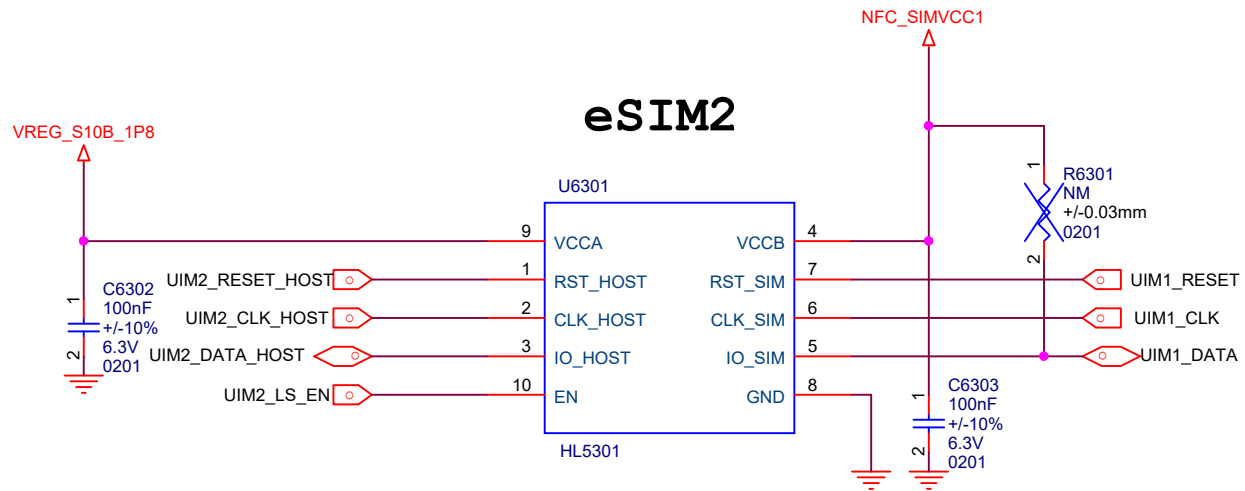
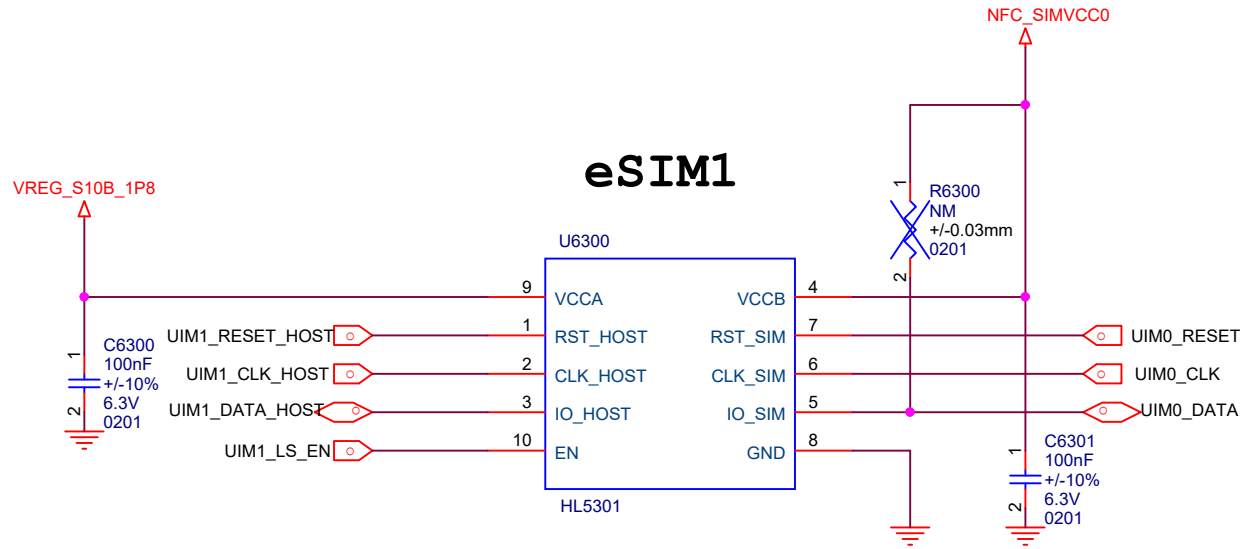
Title		L2_MAIN	Rev P0
Size A3	Document Number		
Date: Thursday, May 05, 2022		Sheet 61 of 79	

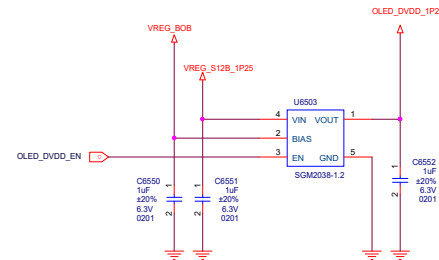
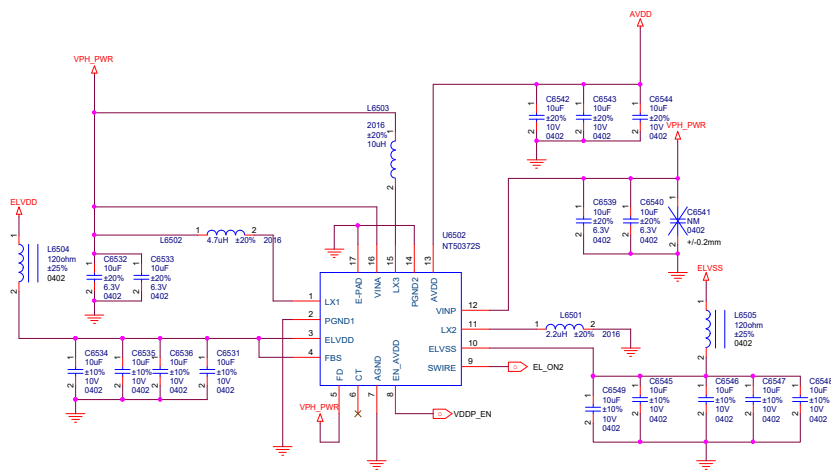
COMPASS

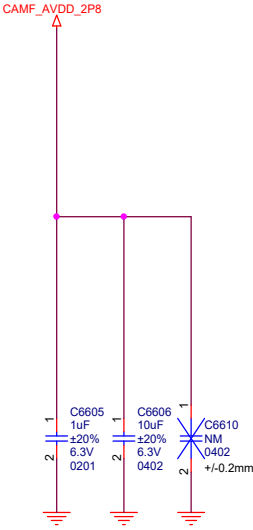


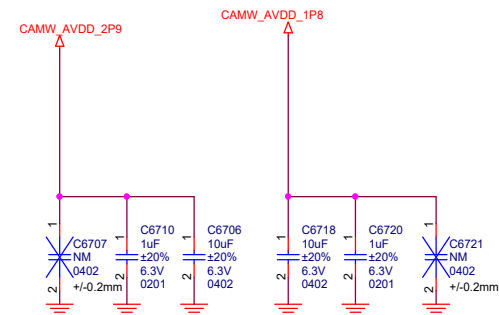
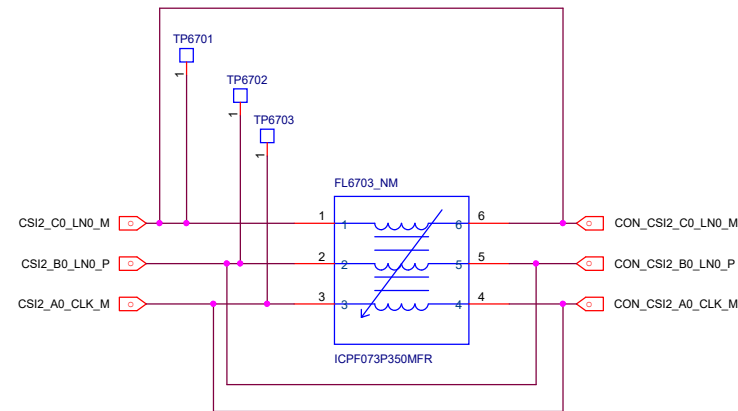
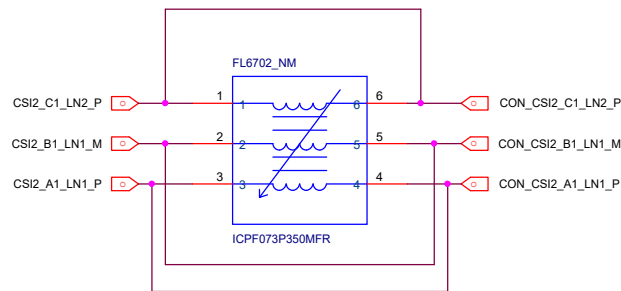
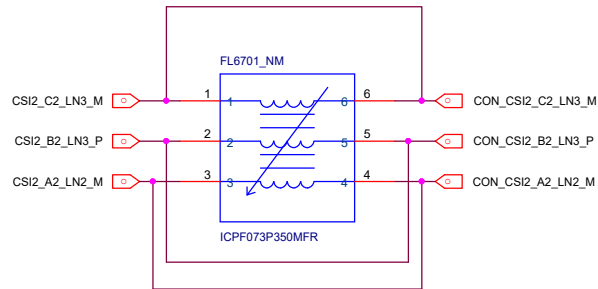
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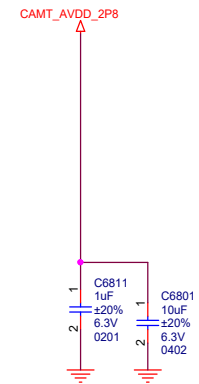
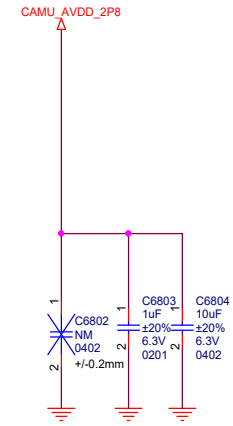


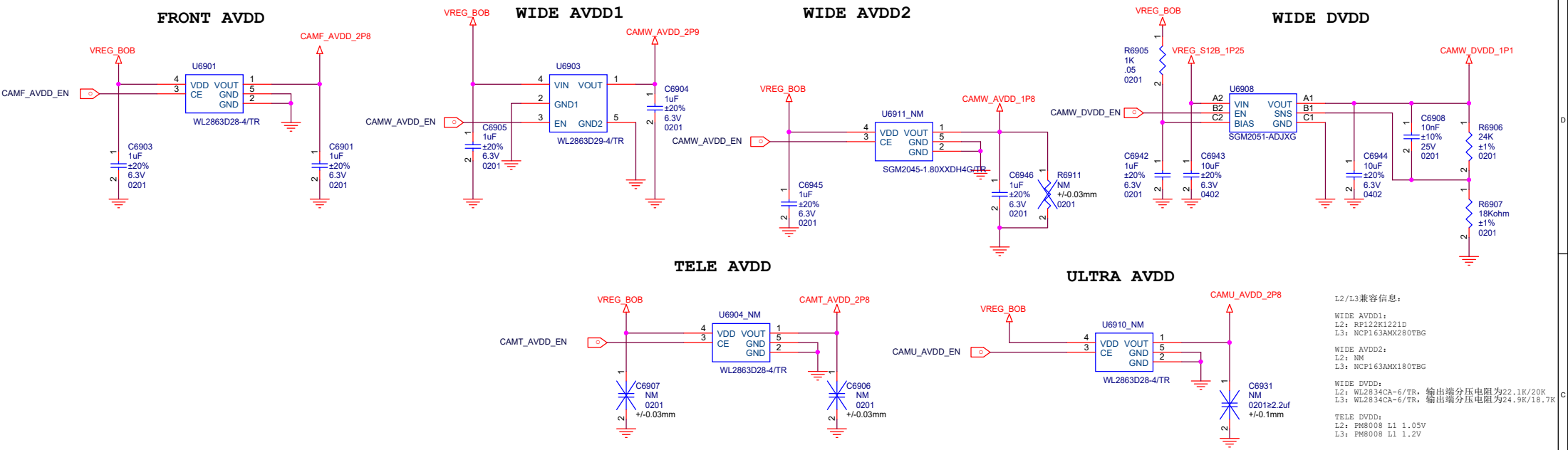




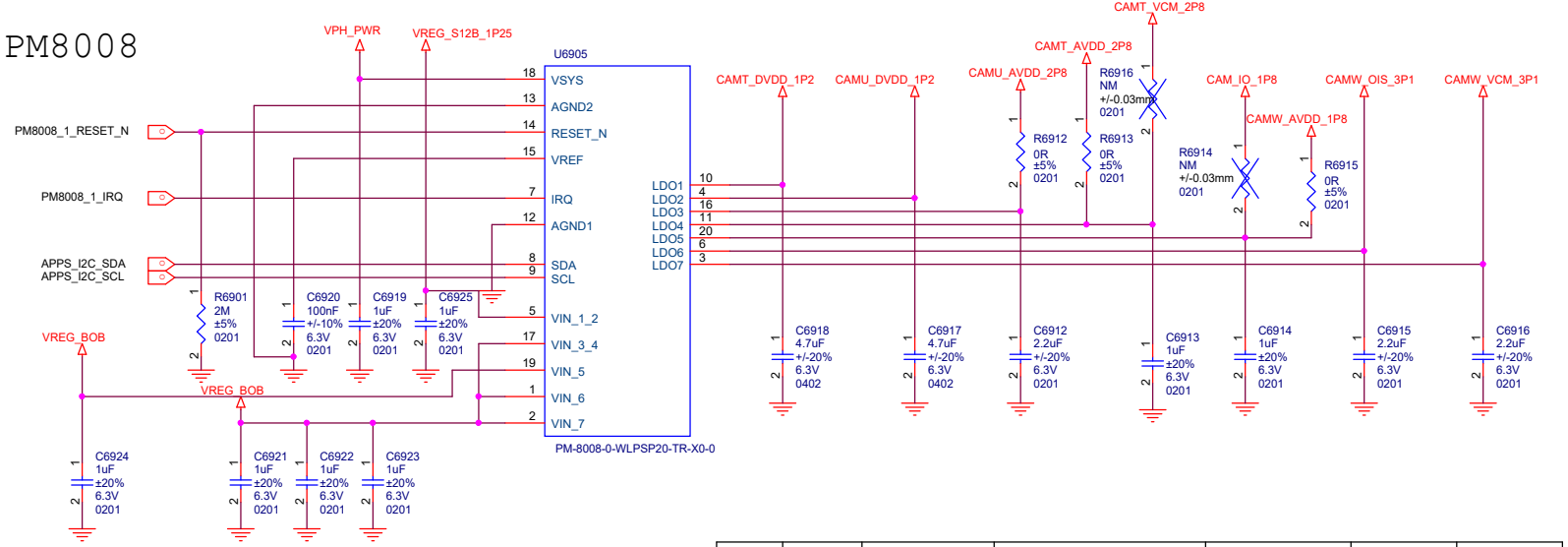




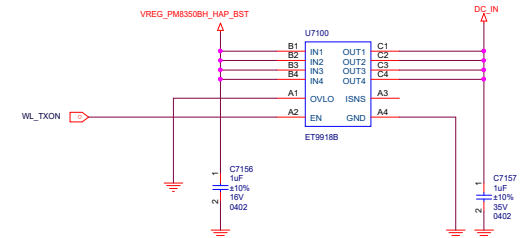
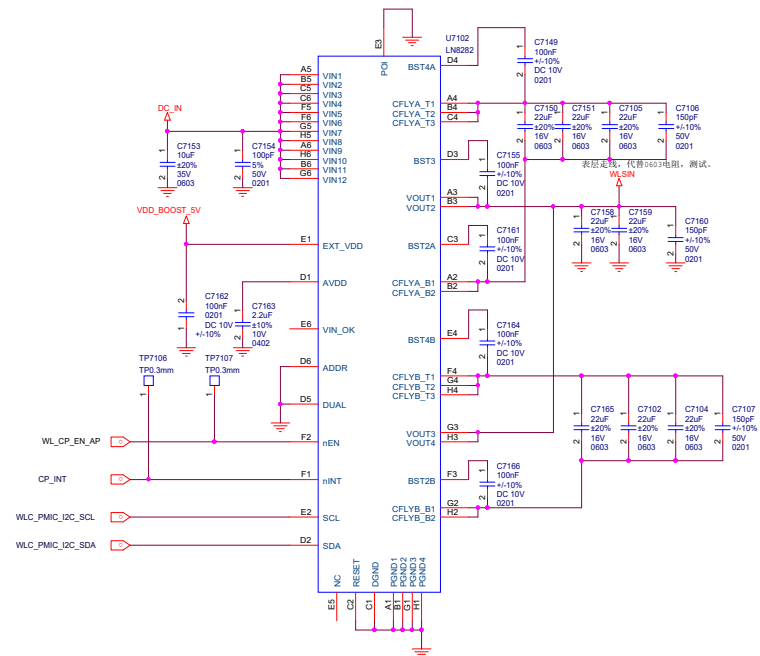
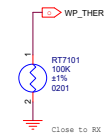
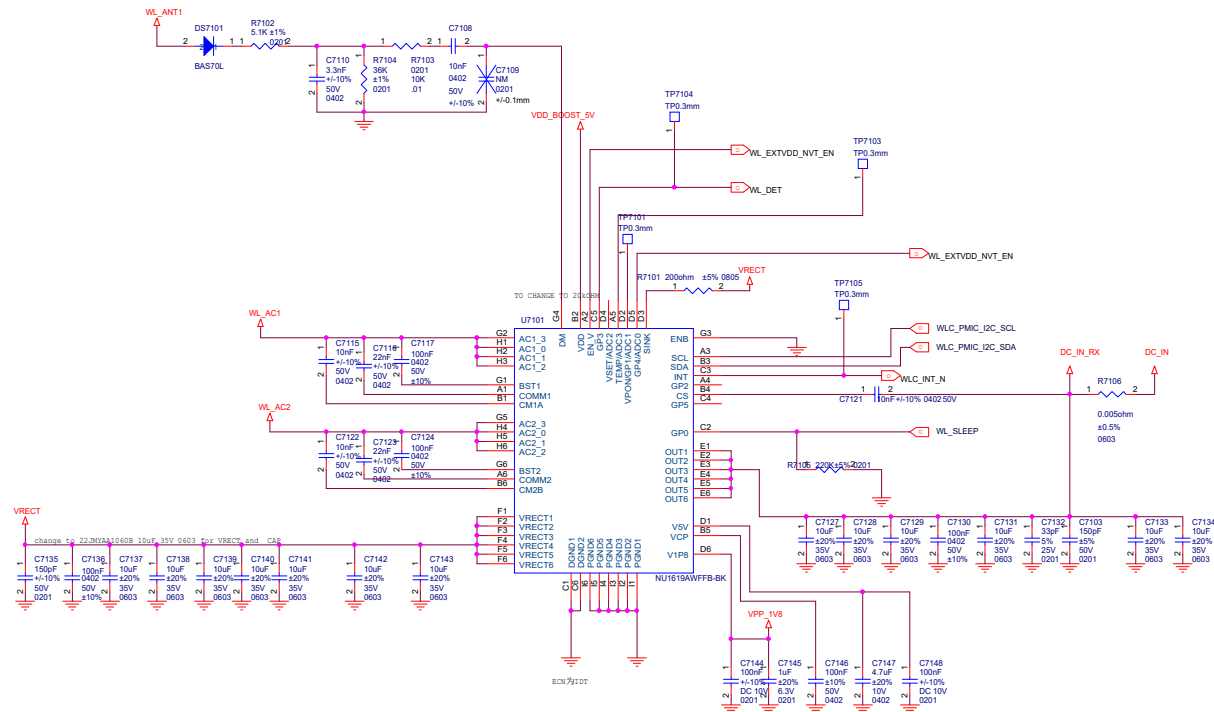


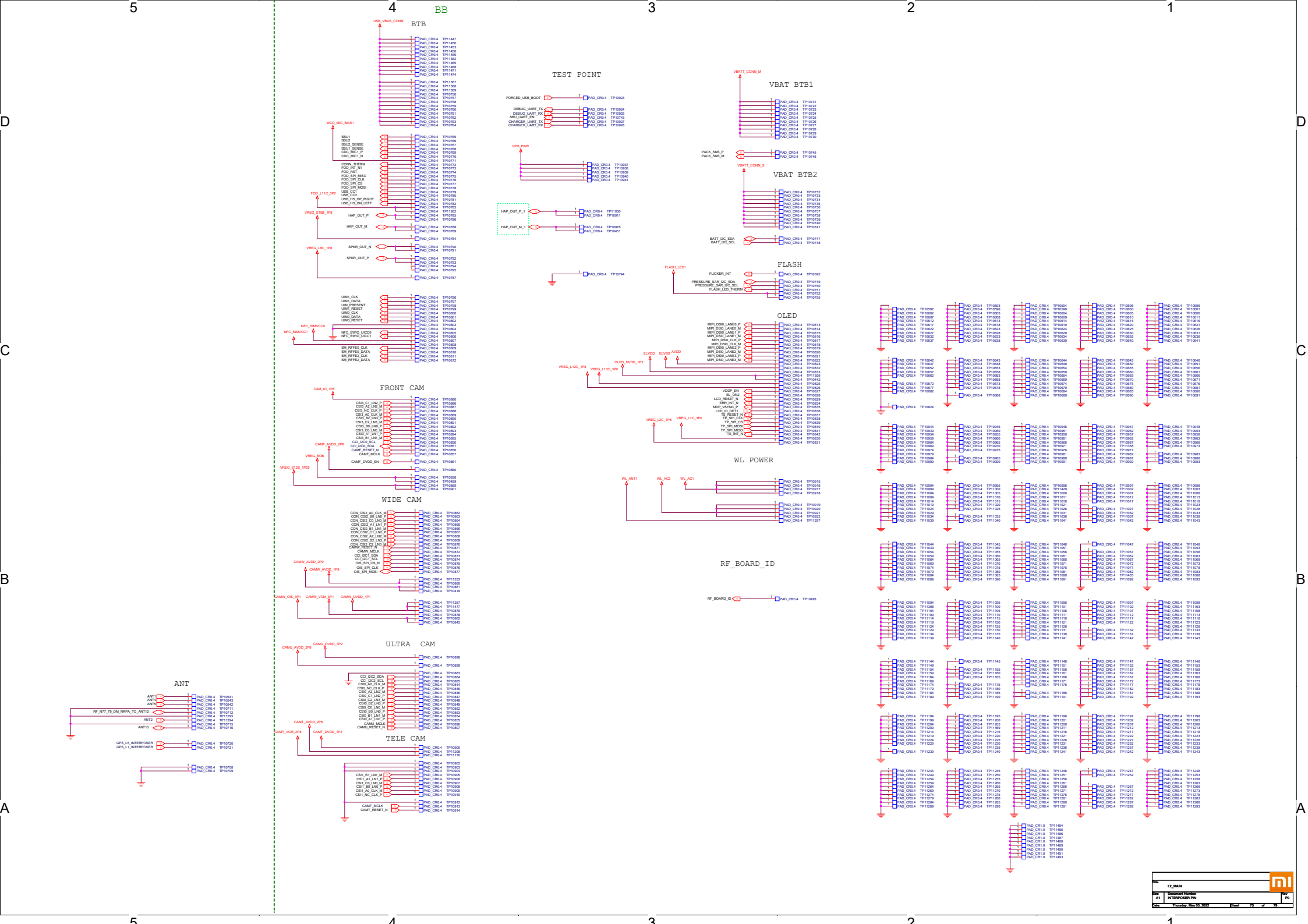


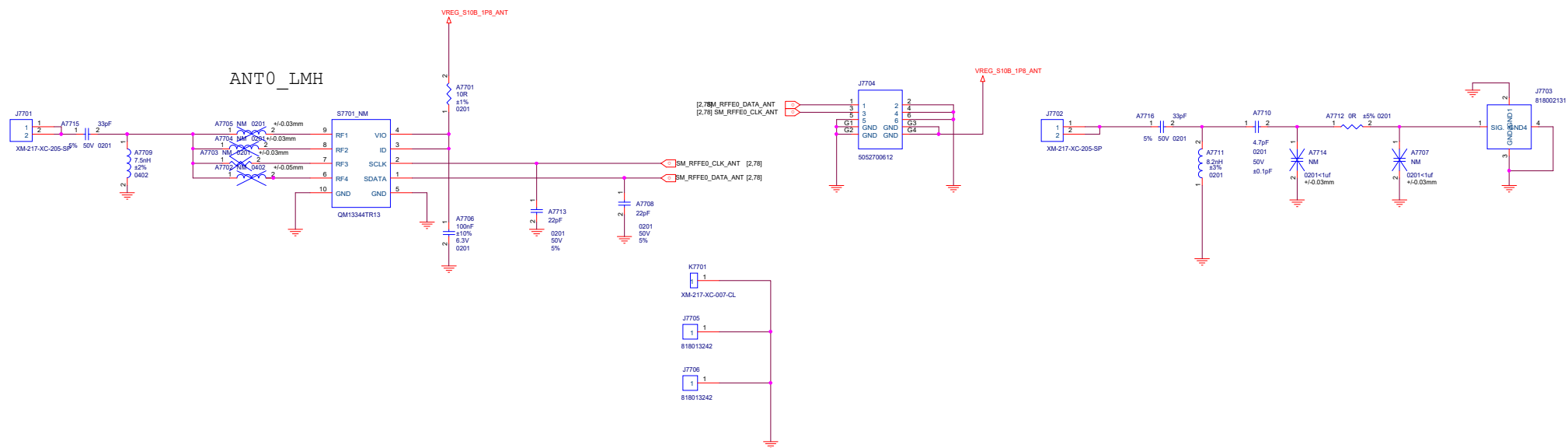
PM8008



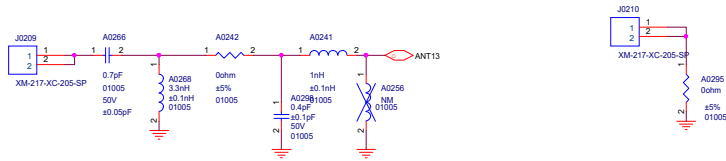
Wireless-POWER



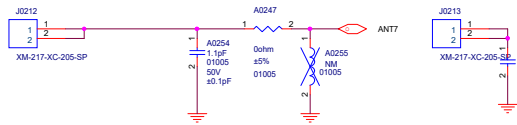


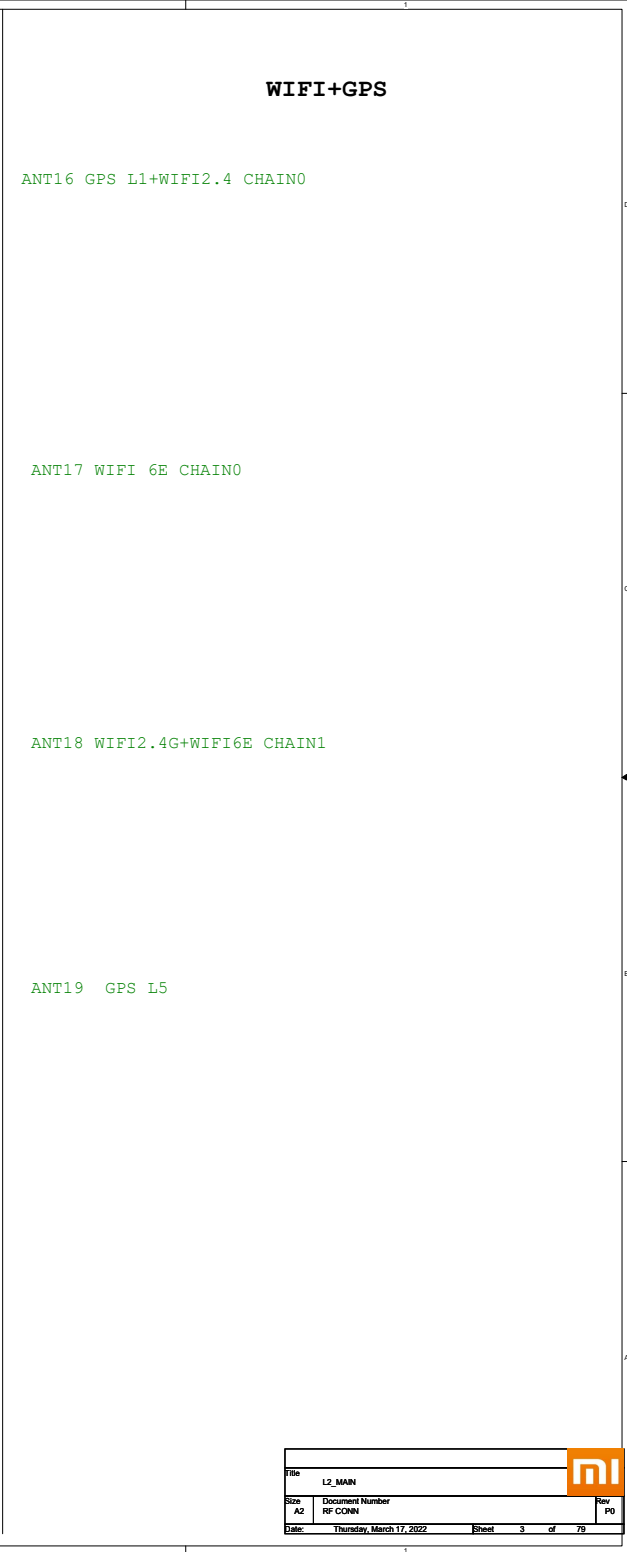
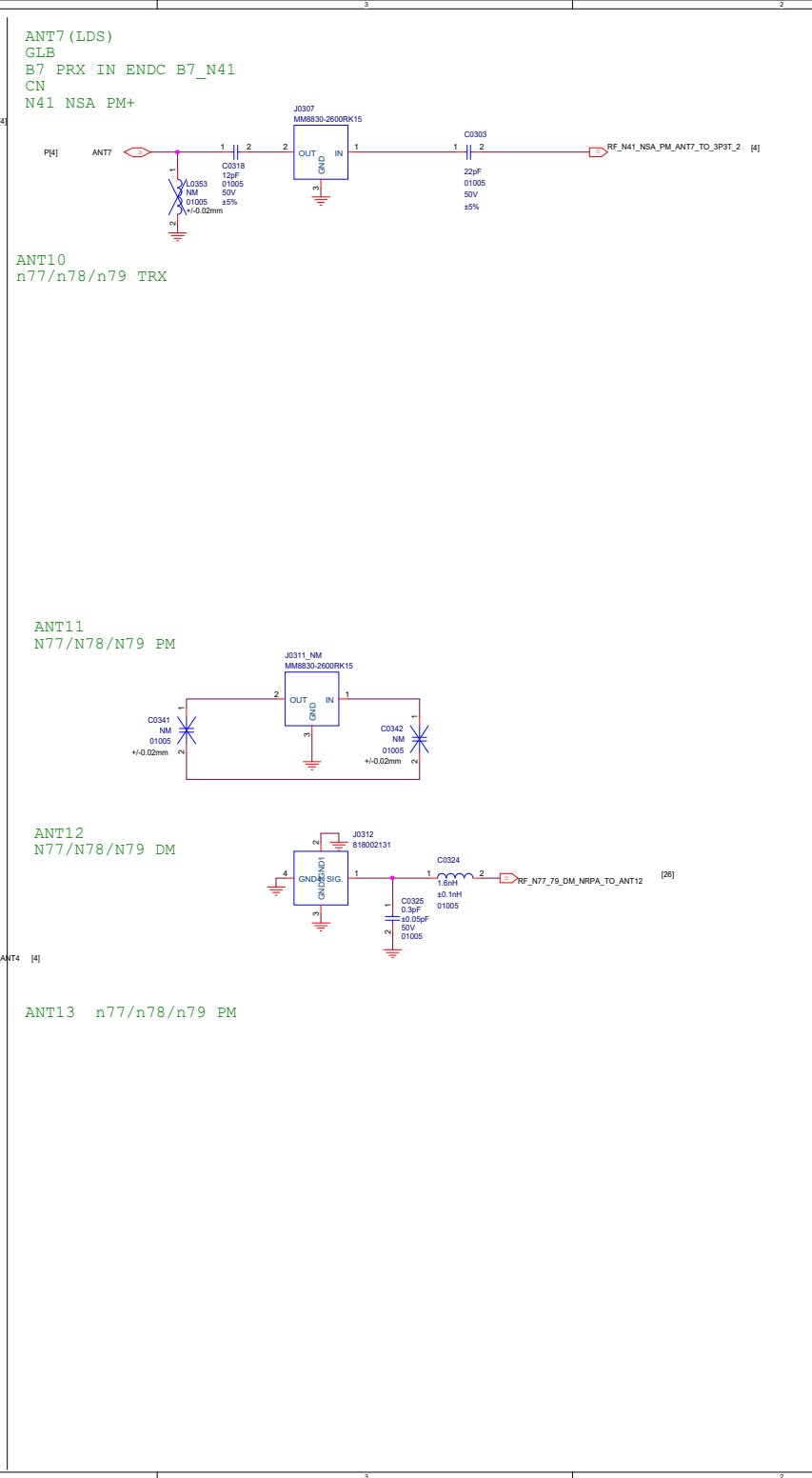
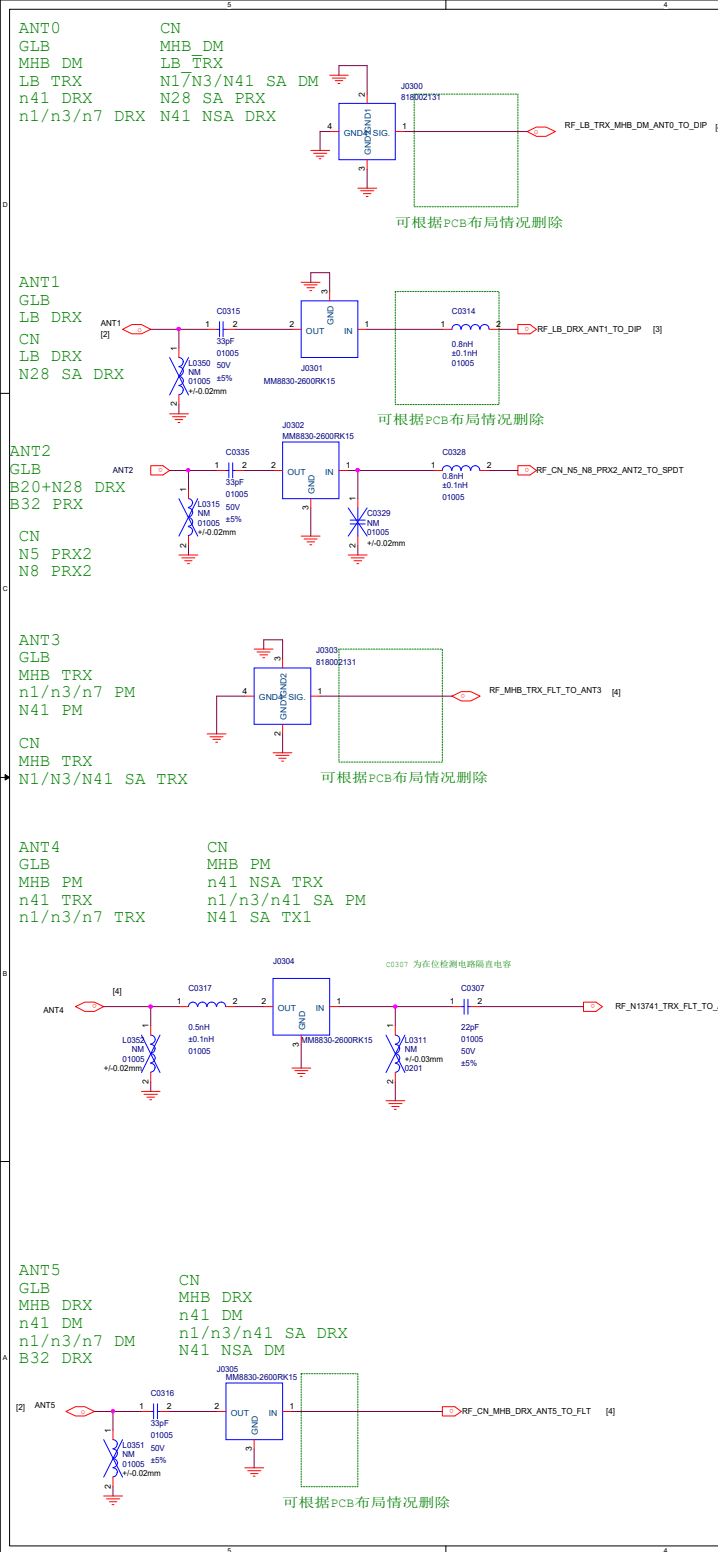


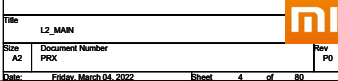
ANT13 N78/N79



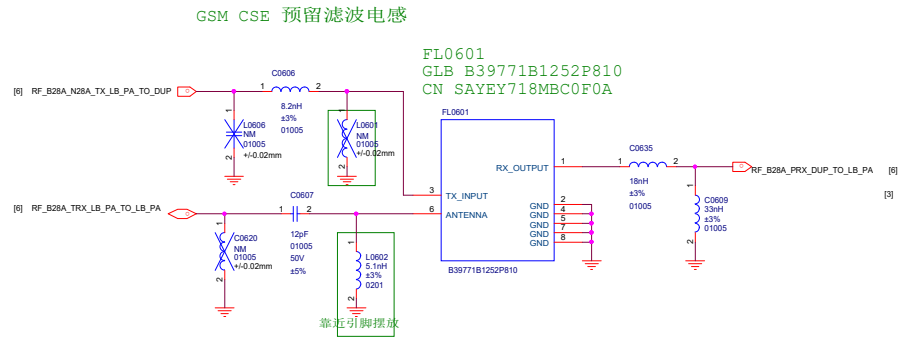
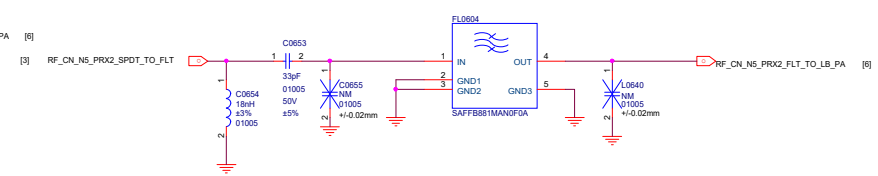
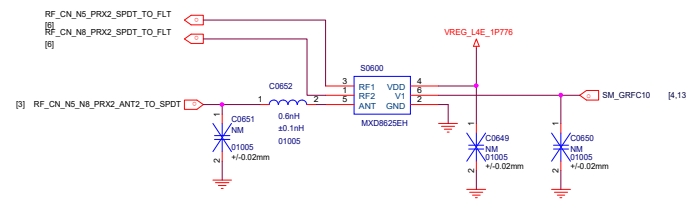
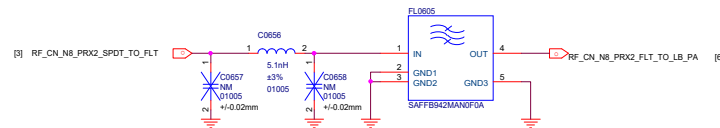
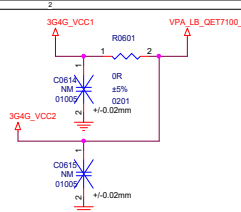
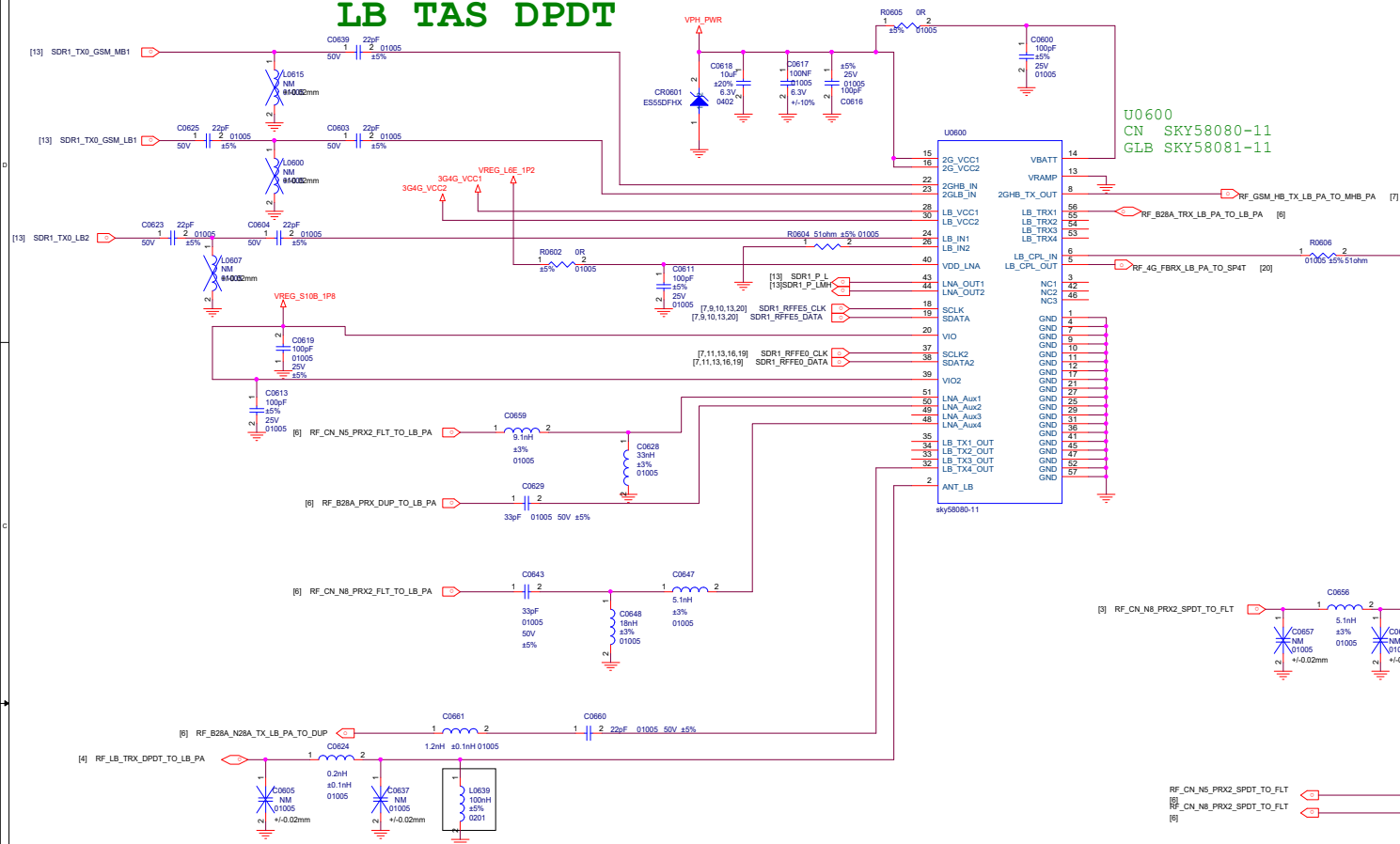
ANT7 N41



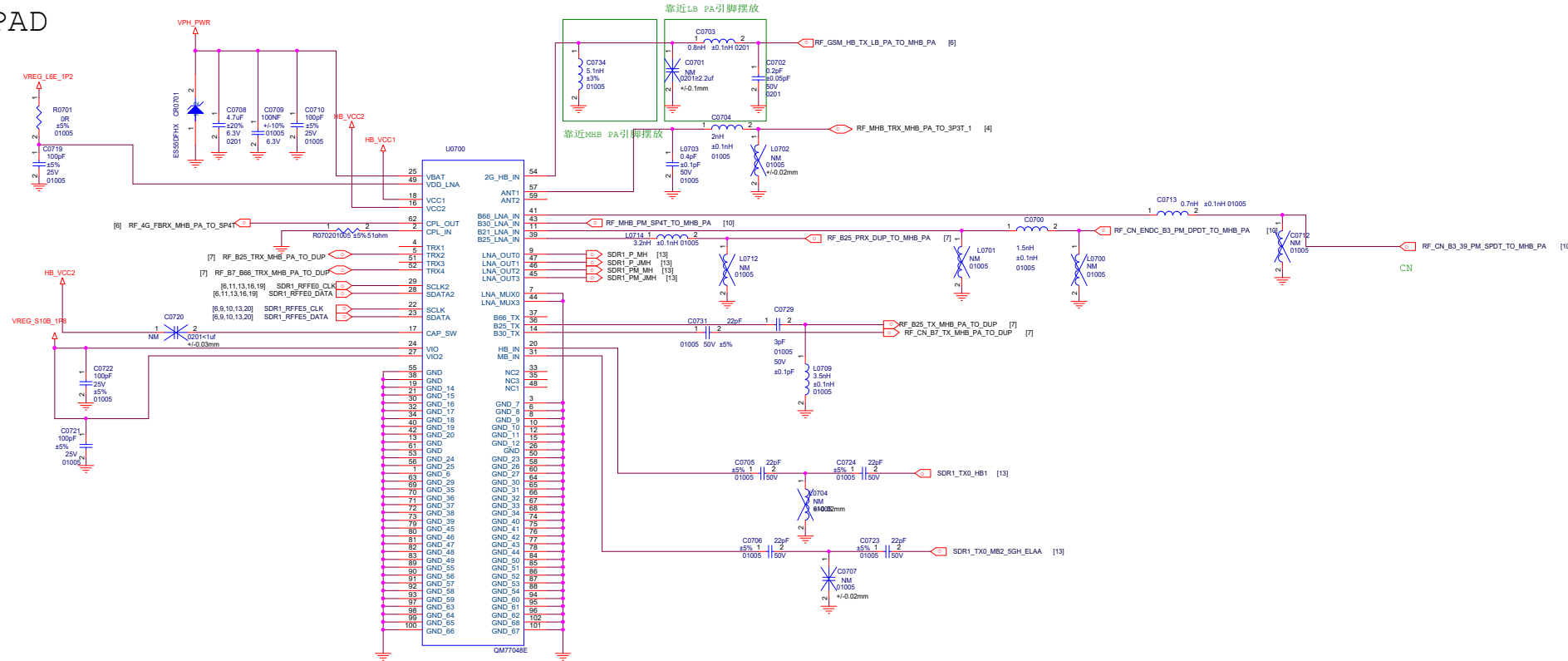




LB TAS DPDT

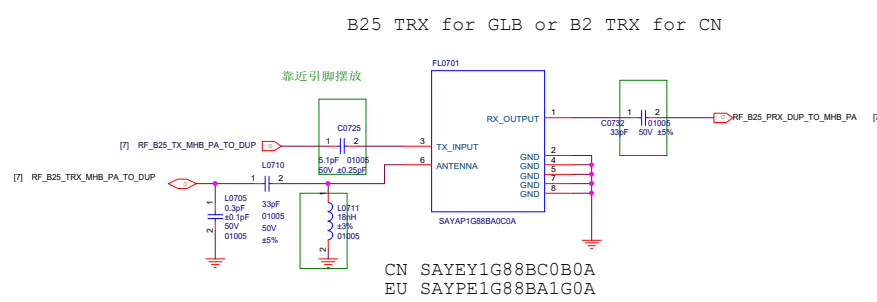


MHB S-PAD

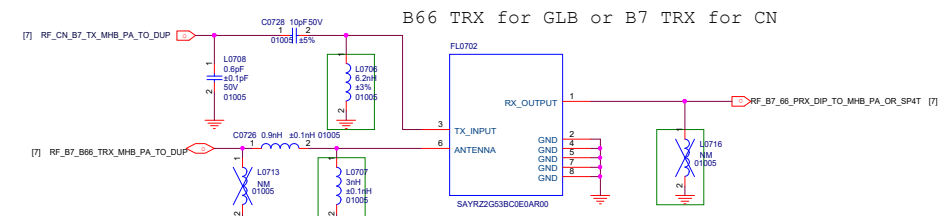


CN QM77048E
EU QM77048B

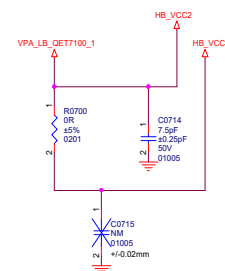
FL0702靠近U0700放置，
兼容走线尽量短

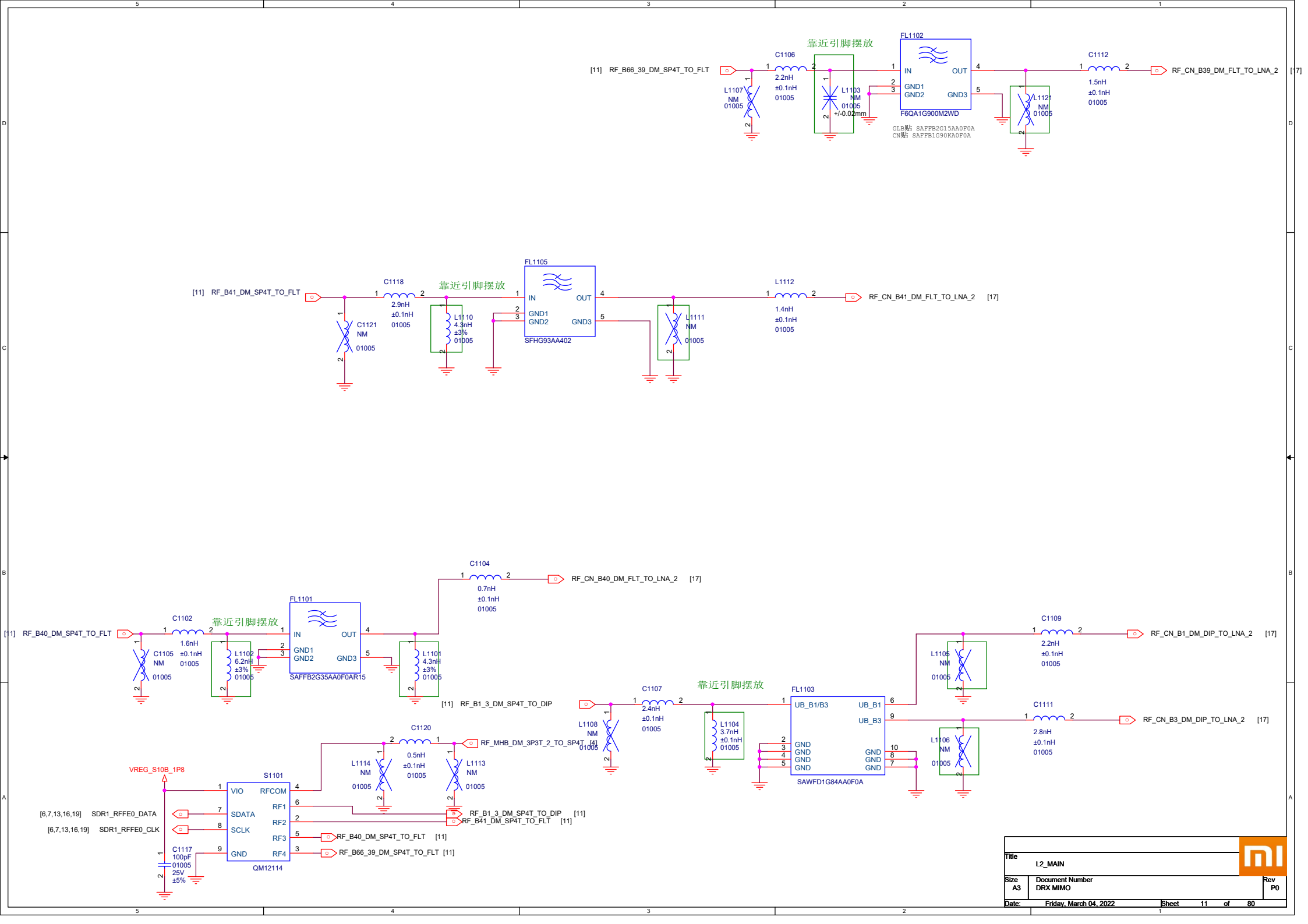


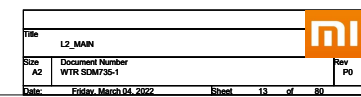
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EU SAYPE1G88BA1G0A

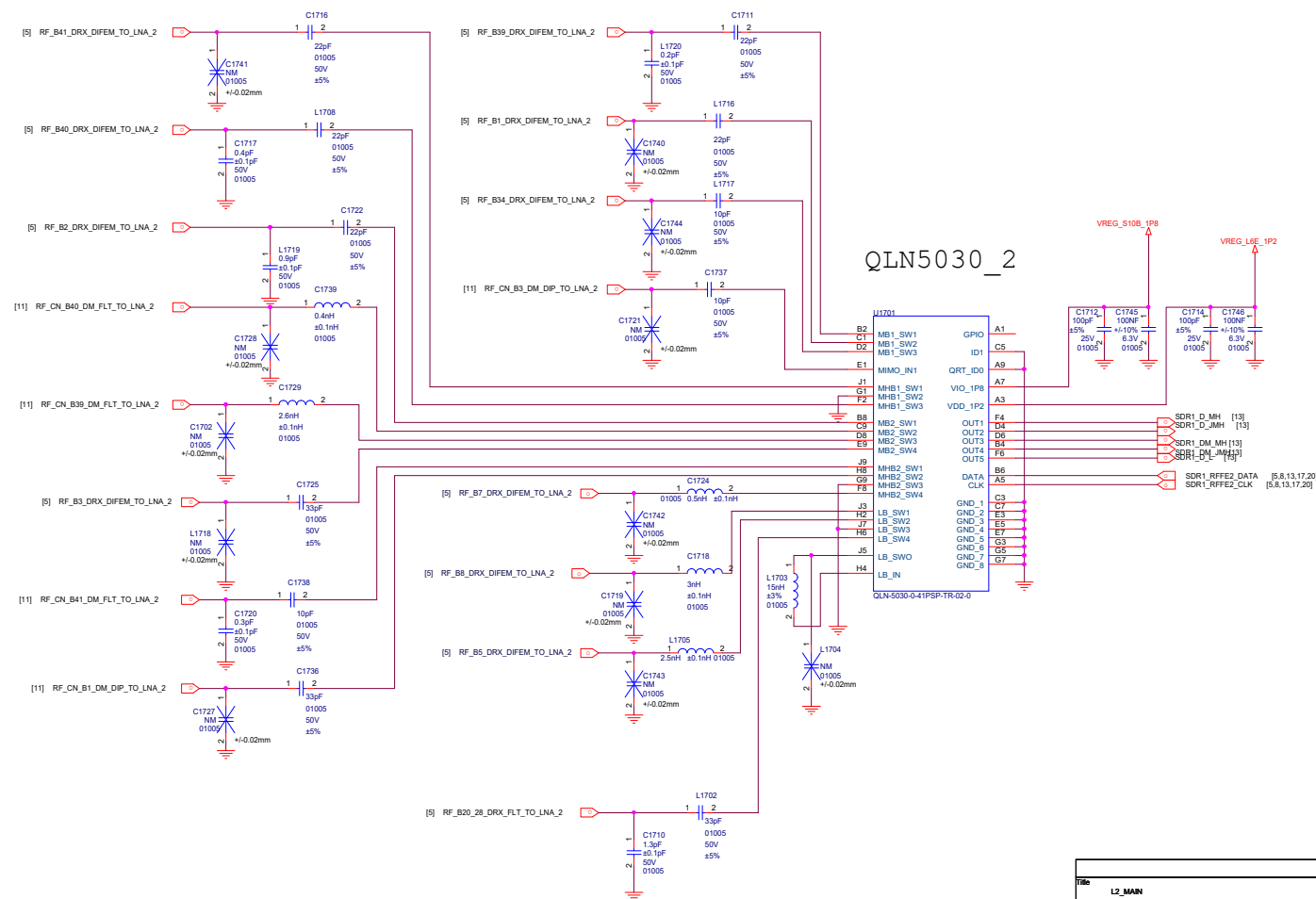


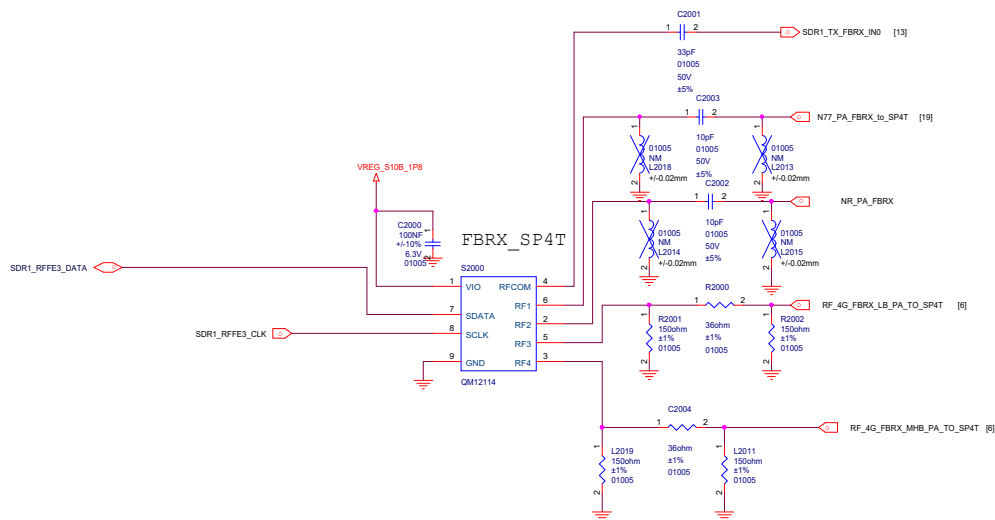
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B1261
GLB B66
B1254

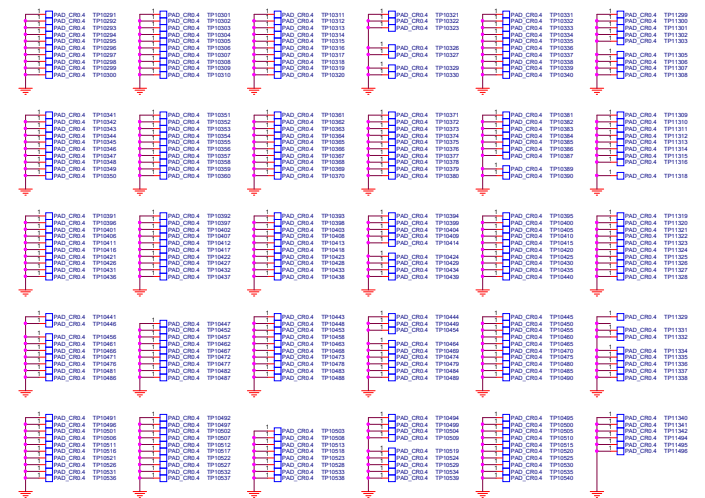


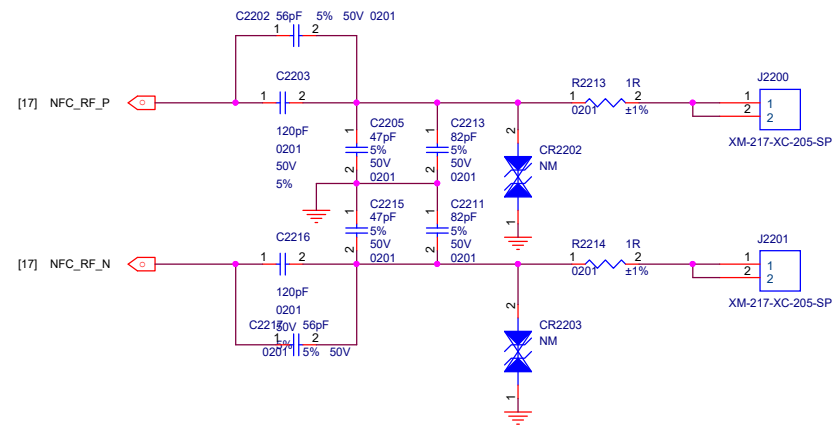


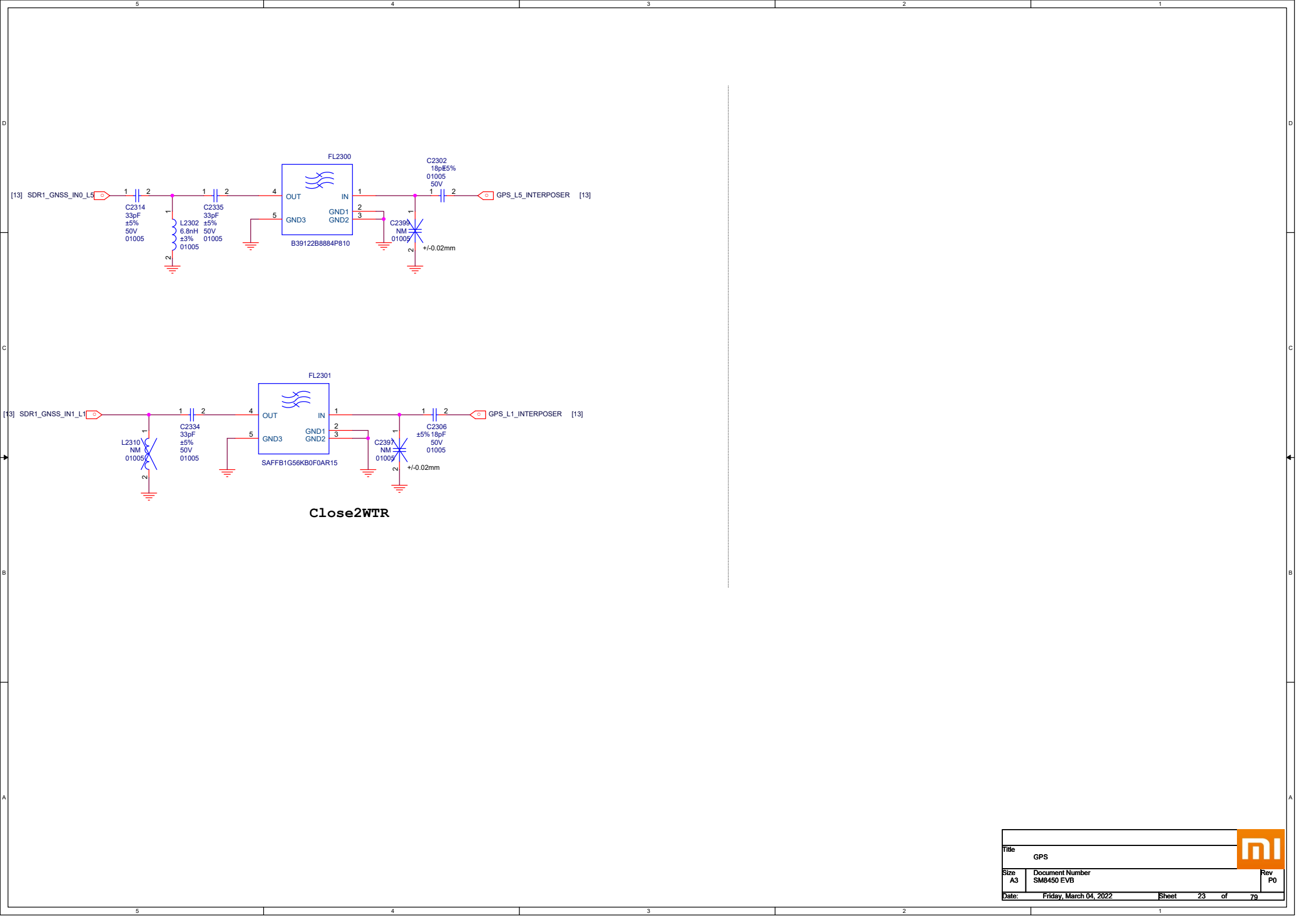


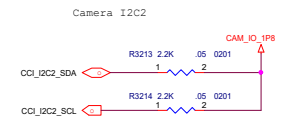
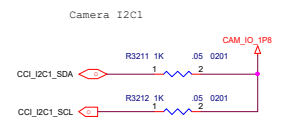
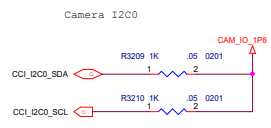




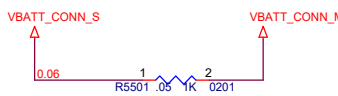
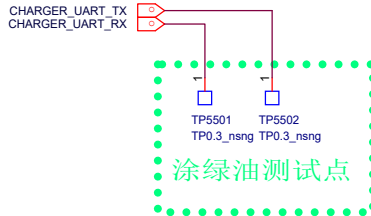
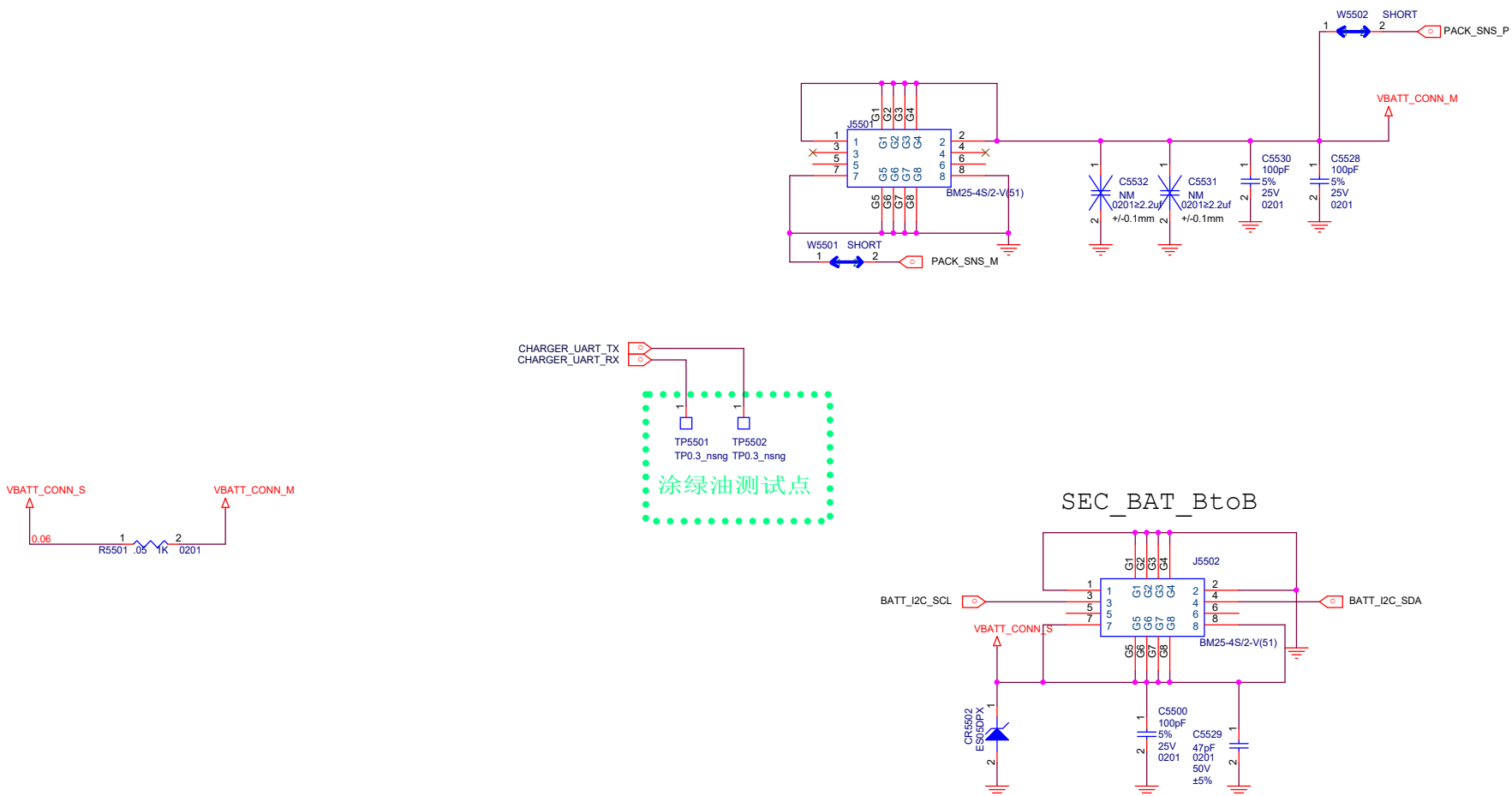


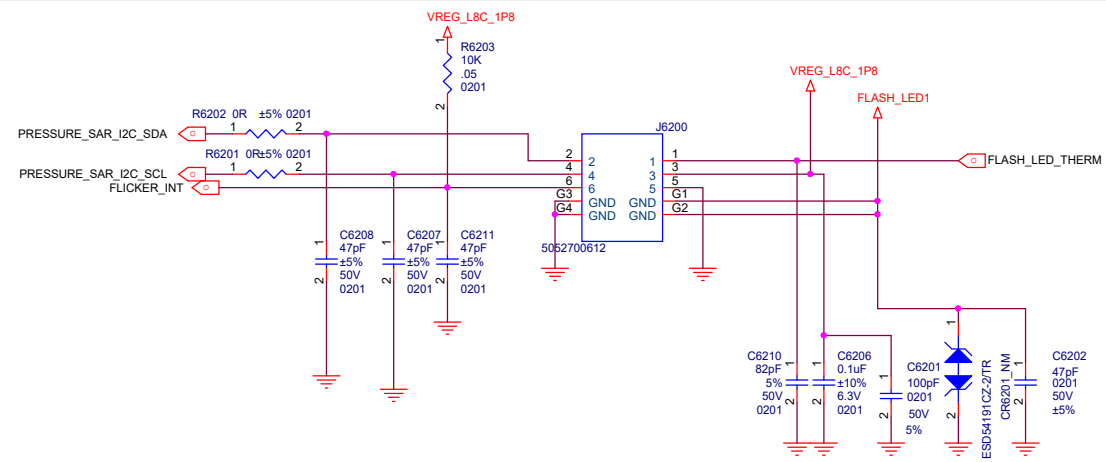


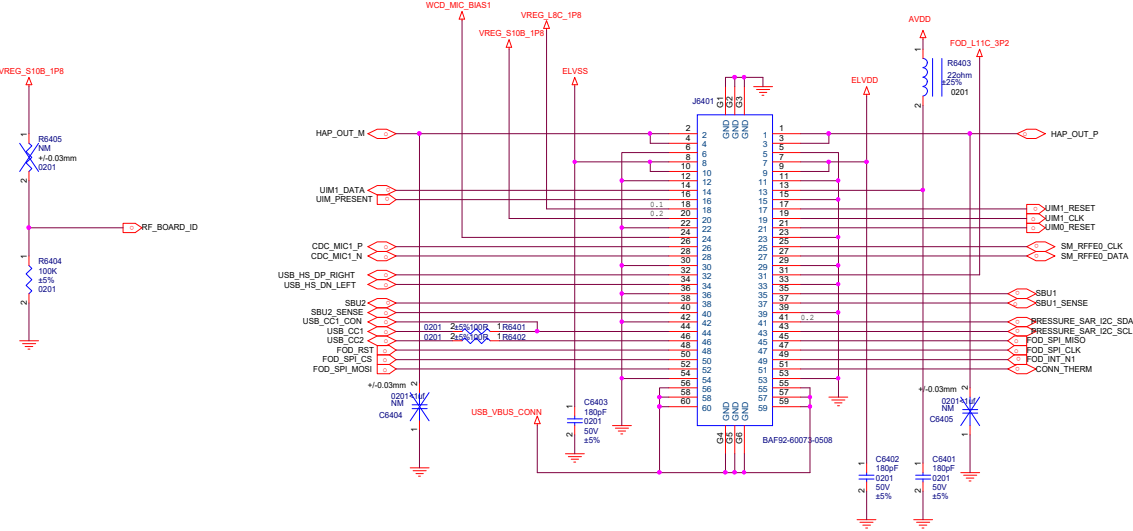




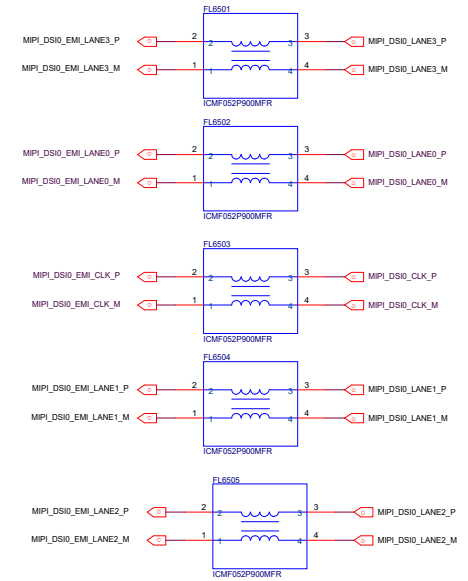
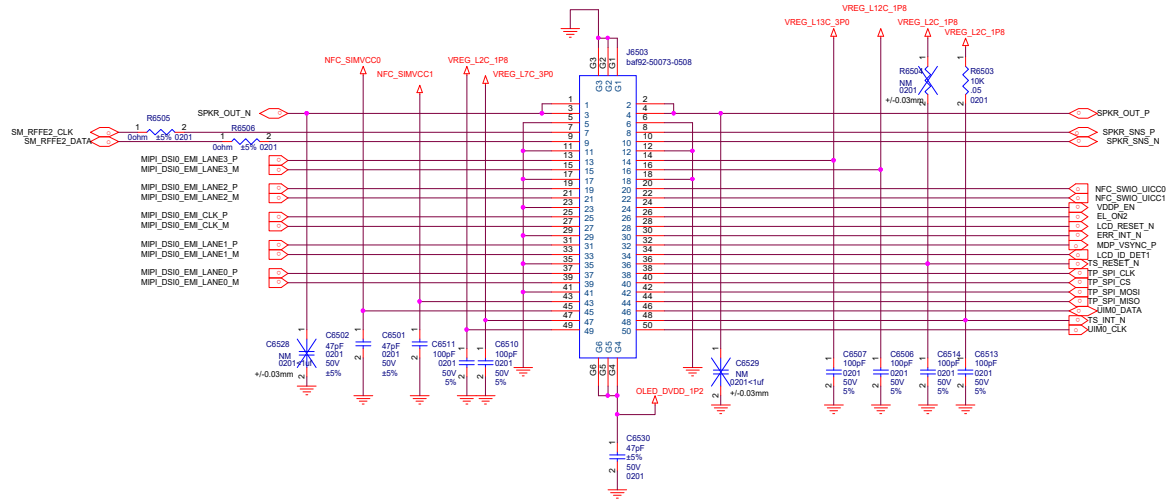
PM8350B---1S BATTERY ,low voltage VIBRATOR DRIVER
PM8350BH---1S BATTERY ,high voltage VIBRATOR DRIVER
PM8350BHS---2S BATTERY ,high voltage VIBRATOR DRIVER

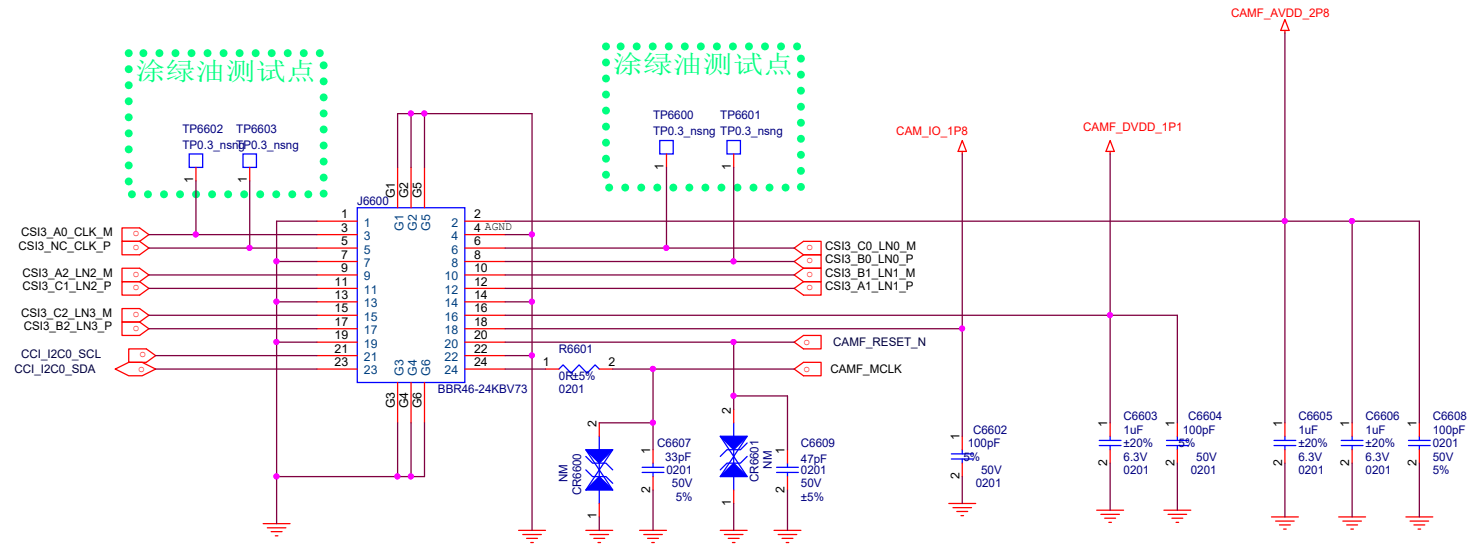


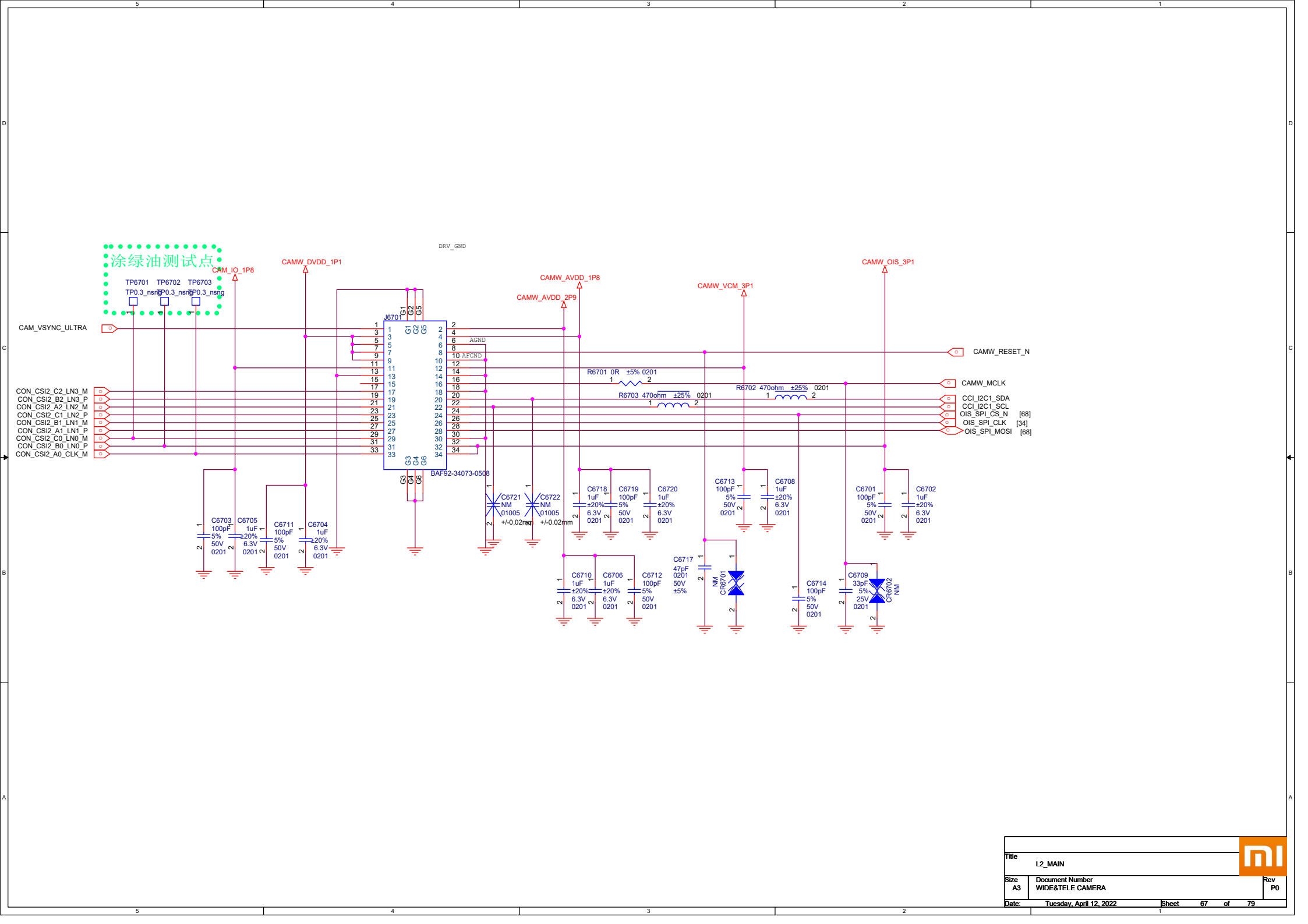




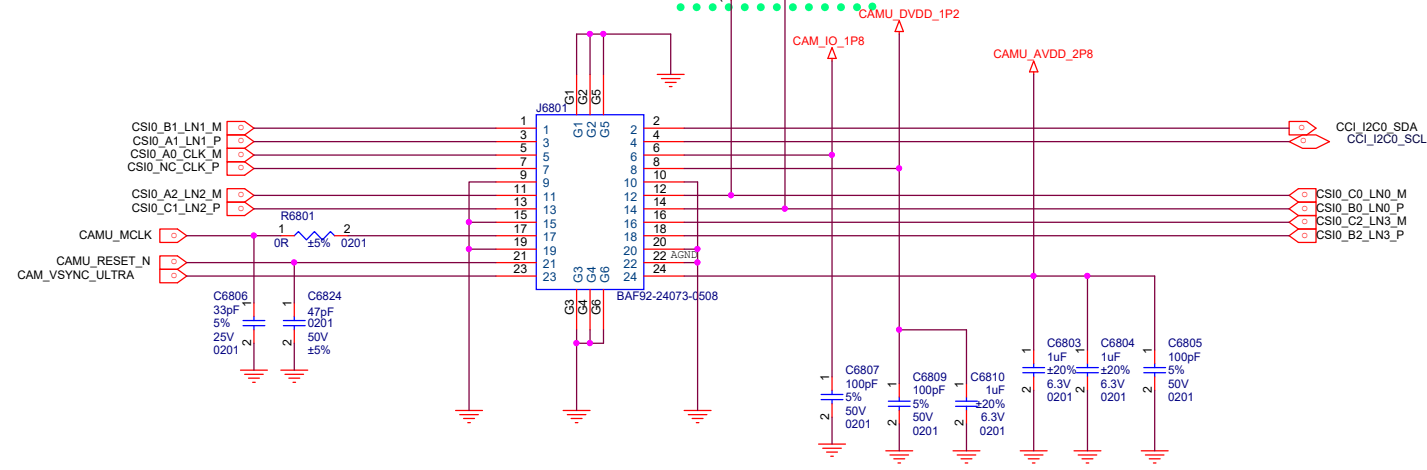
VREG_L2C_1P8: TP1.8
VREG_L7C_3P0: TP3.0
VREG_L10C_1P2: Display 1.2
VREG_L12C_1P8: Display 1.8
VREG_L13C_3P0: Display 3.0



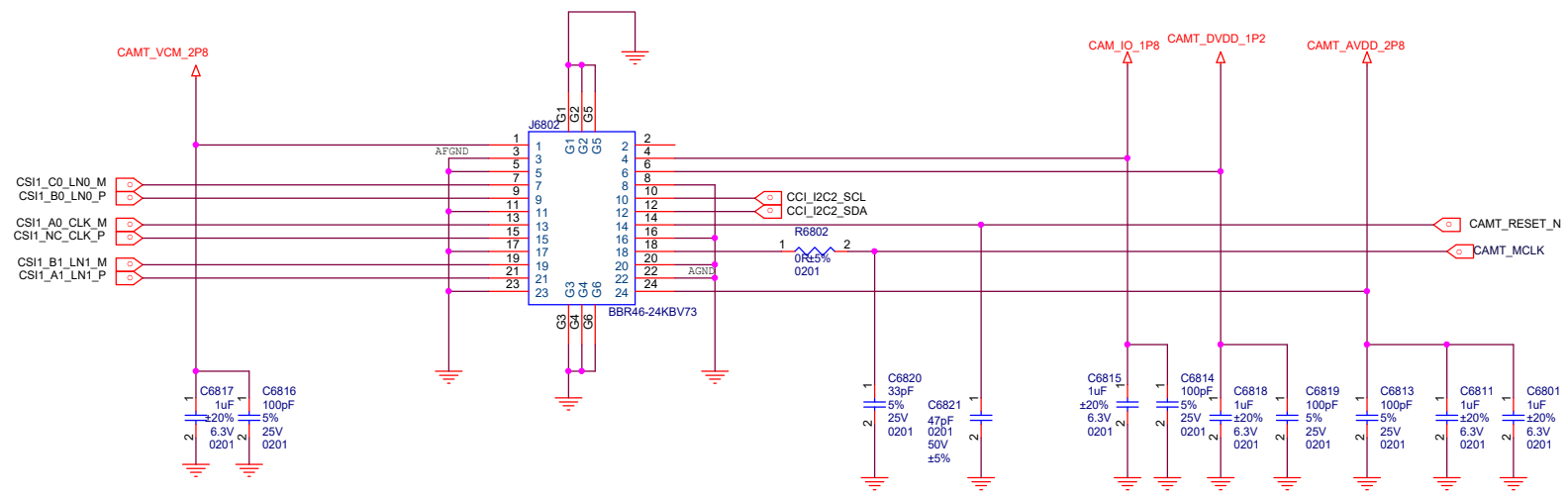


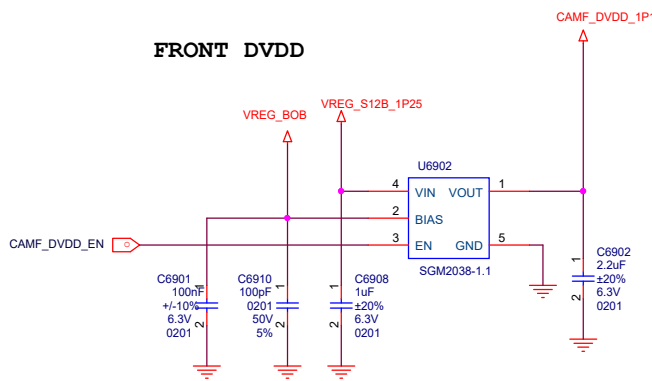


ULTRA



TELE





L2/L3兼容信息:

WIDE AVDD1:
L2: RP122K1221D
L3: NCP163AMX280TBG

WIDE AVDD2:
L2: NM
L3: NCP163AMX180TBG

WIDE DVDD:
L2: WL2834CA-6/TR, 输出端分压电阻为22.1K/20K
L3: WL2834CA-6/TR, 输出端分压电阻为24.9K/18.7K

TELE DVDD:
L2: PM8008 L1 1.05V
L3: PM8008 L1 1.2V

LDO	Irate (ma)	Default On Voltage(V)	Working Voltage			Net Name	Capless	Cout(uf)
			Min	Normal	Max			
L1	1000	-- --	0.95	1.05	1.15	Tele DVDD	NO	4.7~20
L2	1000	-- --	1.14	1.2	1.26	Ultr DVDD	NO	4.7~20
L3	400	-- --	2.7	2.8	2.9	Ultr AVDD(兼容)	YES	1.0~50
L4	400	-- --	2.7	2.8	-- --	Tele VCM	YES	1.0~50
L5	400	-- --	1.7	1.8	1.9	Cam IOVDD	YES	1.0~50
L6	400	-- --	-- --	3.1	-- --	Wide OIS	YES	1.0~50
L7	400	-- --	-- --	3.1	-- --	Wide VCM	YES	1.0~50

Title

L2_MAIN

Size

A3

Document Number

PM8008 & CAMERA LDOs

Date:

Friday, March 04, 2022

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D

C

B

A

